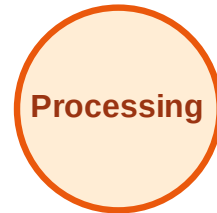


BFS Traversal

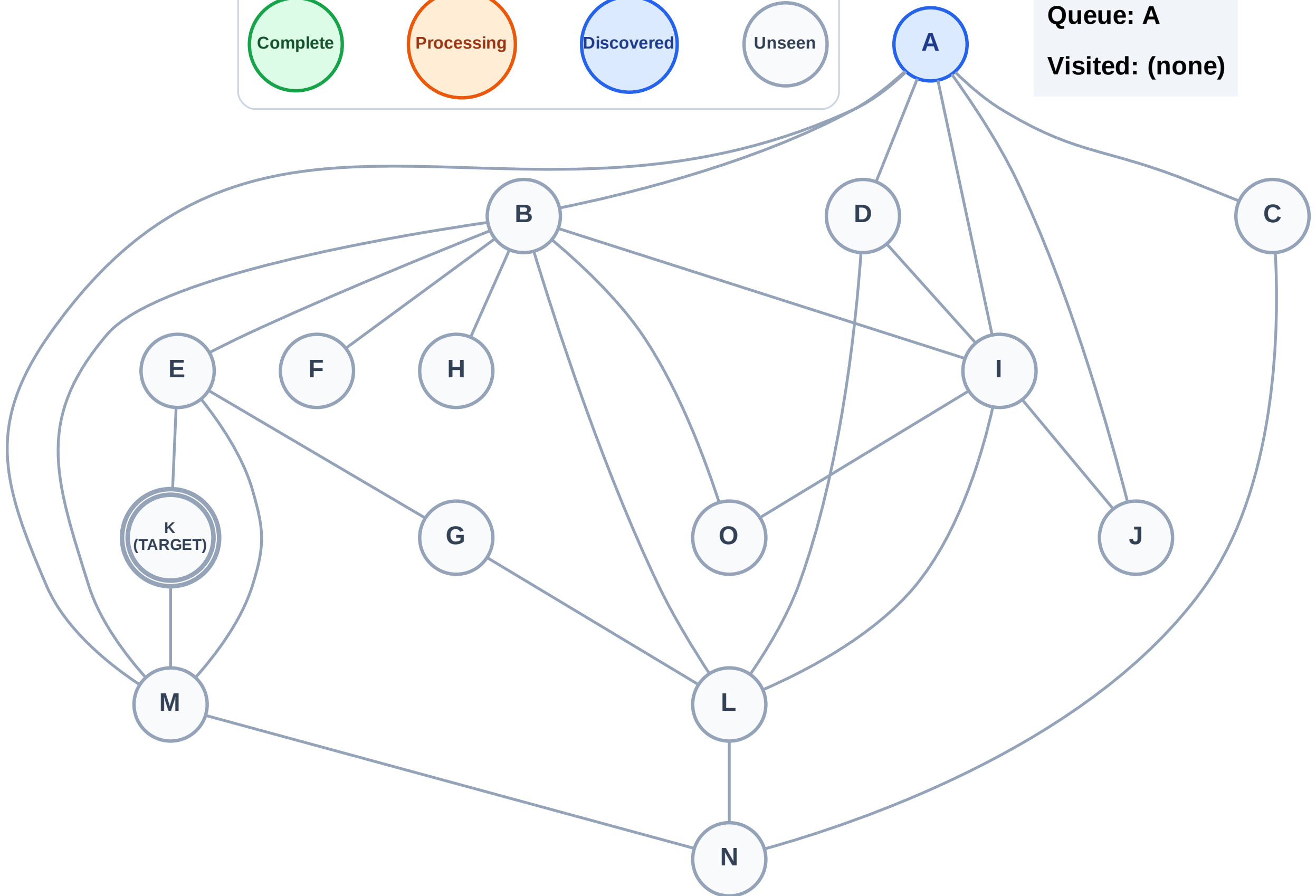
Step 1: Start at A (Seeking target node K)

Legend



Queue: A

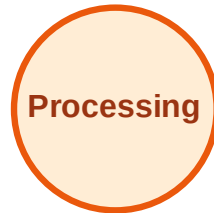
Visited: (none)



BFS Traversal

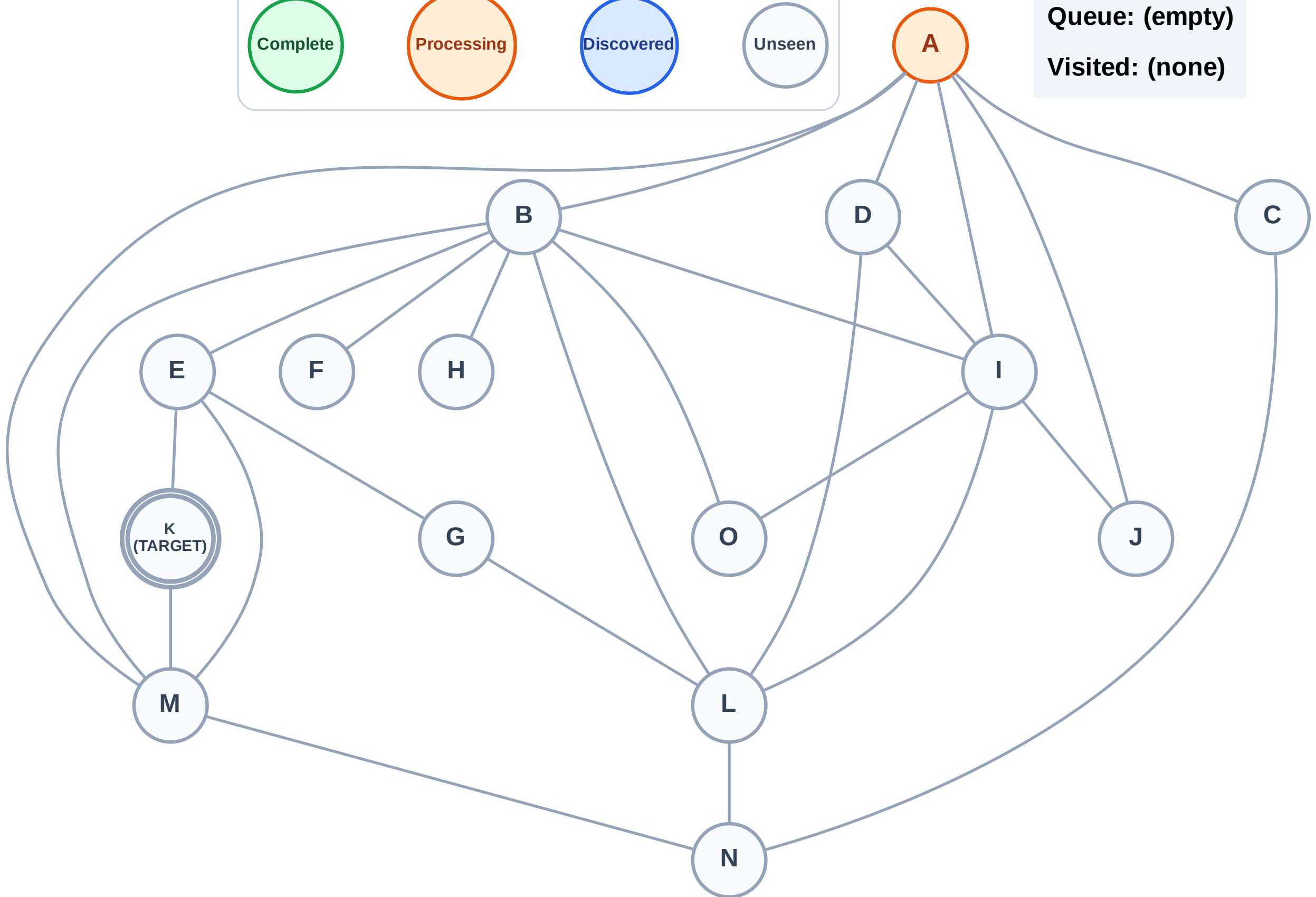
Step 2: Process node A

Legend



Queue: (empty)

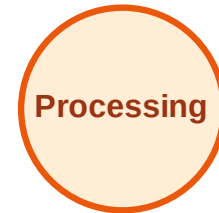
Visited: (none)



BFS Traversal

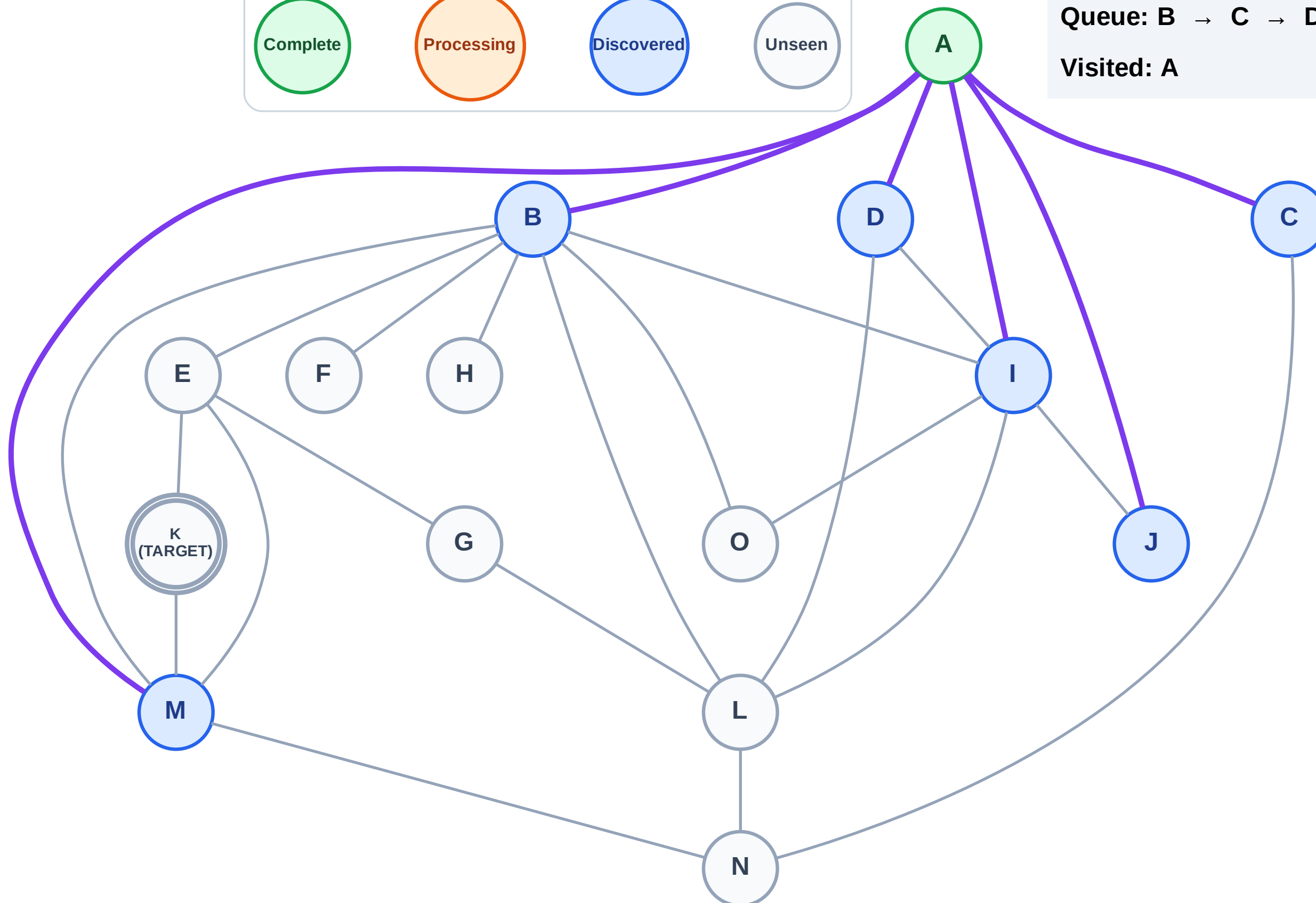
Step 3: Discover B, C, D, I, J, M from A

Legend



Queue: B → C → D → I → J → M

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

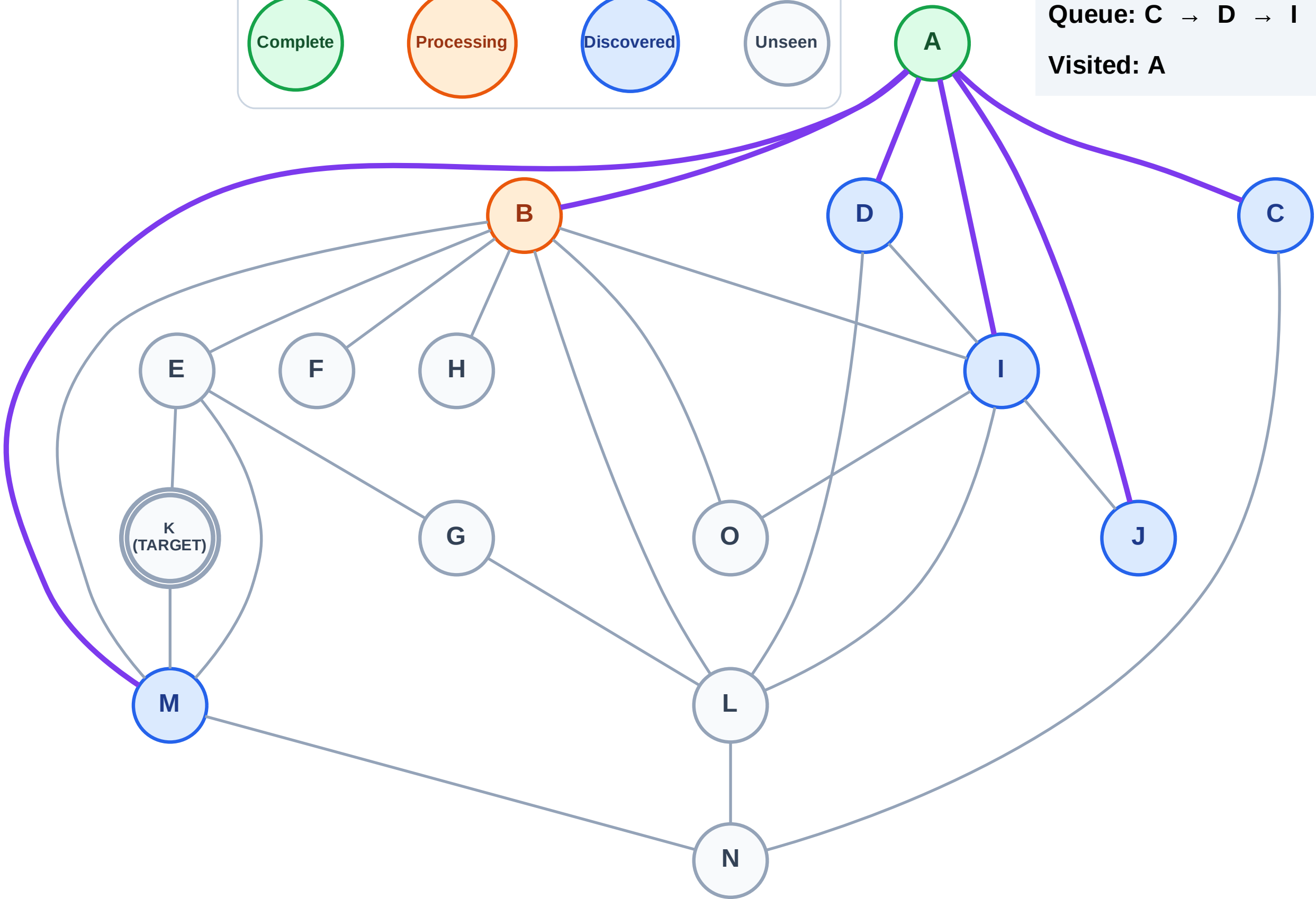
Processing

Discovered

Unseen

Queue: C → D → I → J → M

Visited: A



BFS Traversal

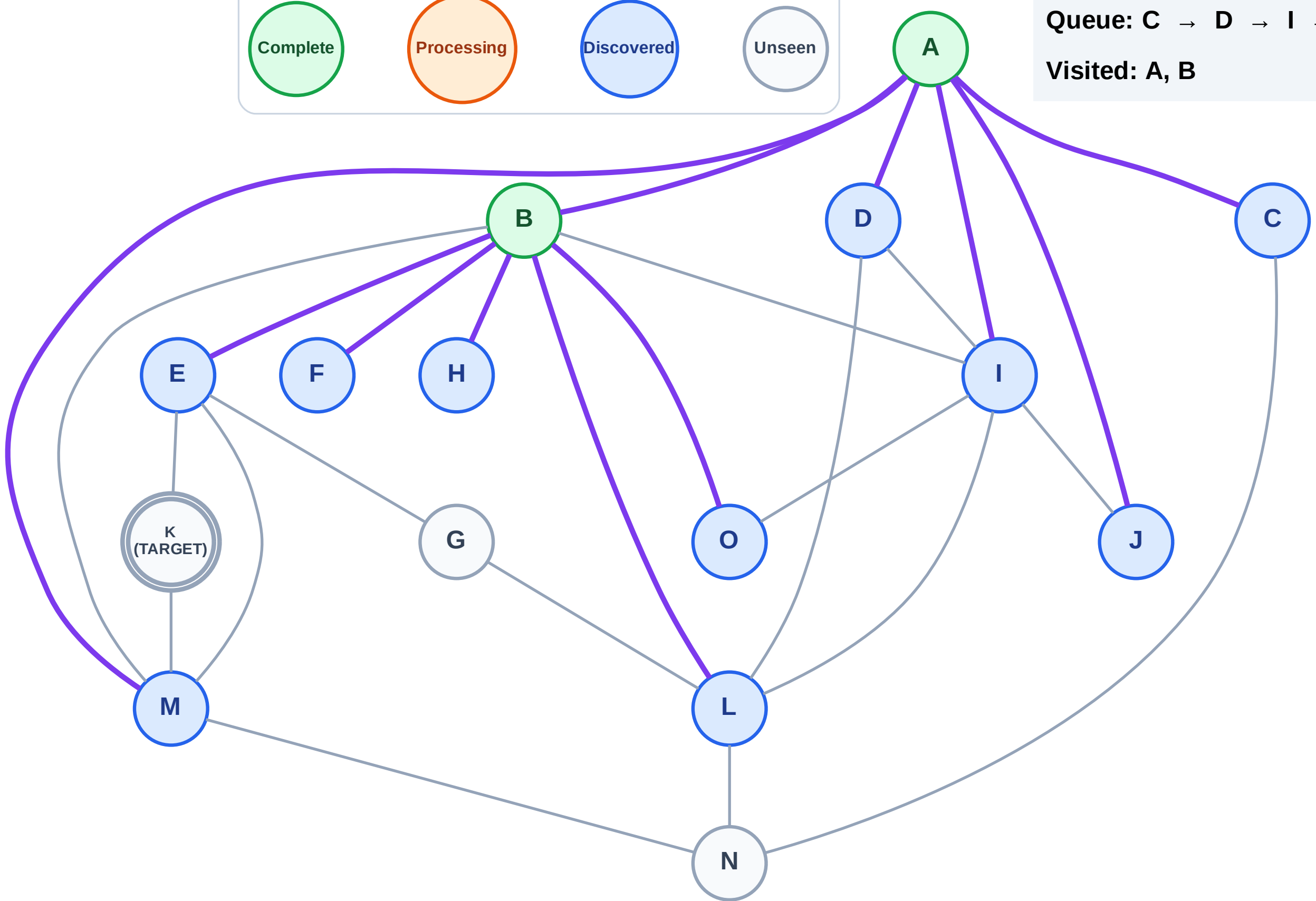
Step 5: Discover E, F, H, L, O from B

Legend



Queue: C → D → I → J → M → E → F → H → L → O

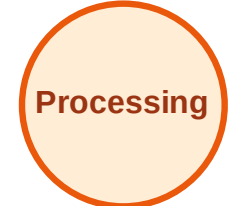
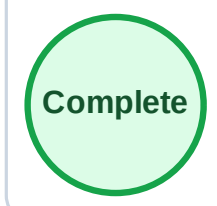
Visited: A, B



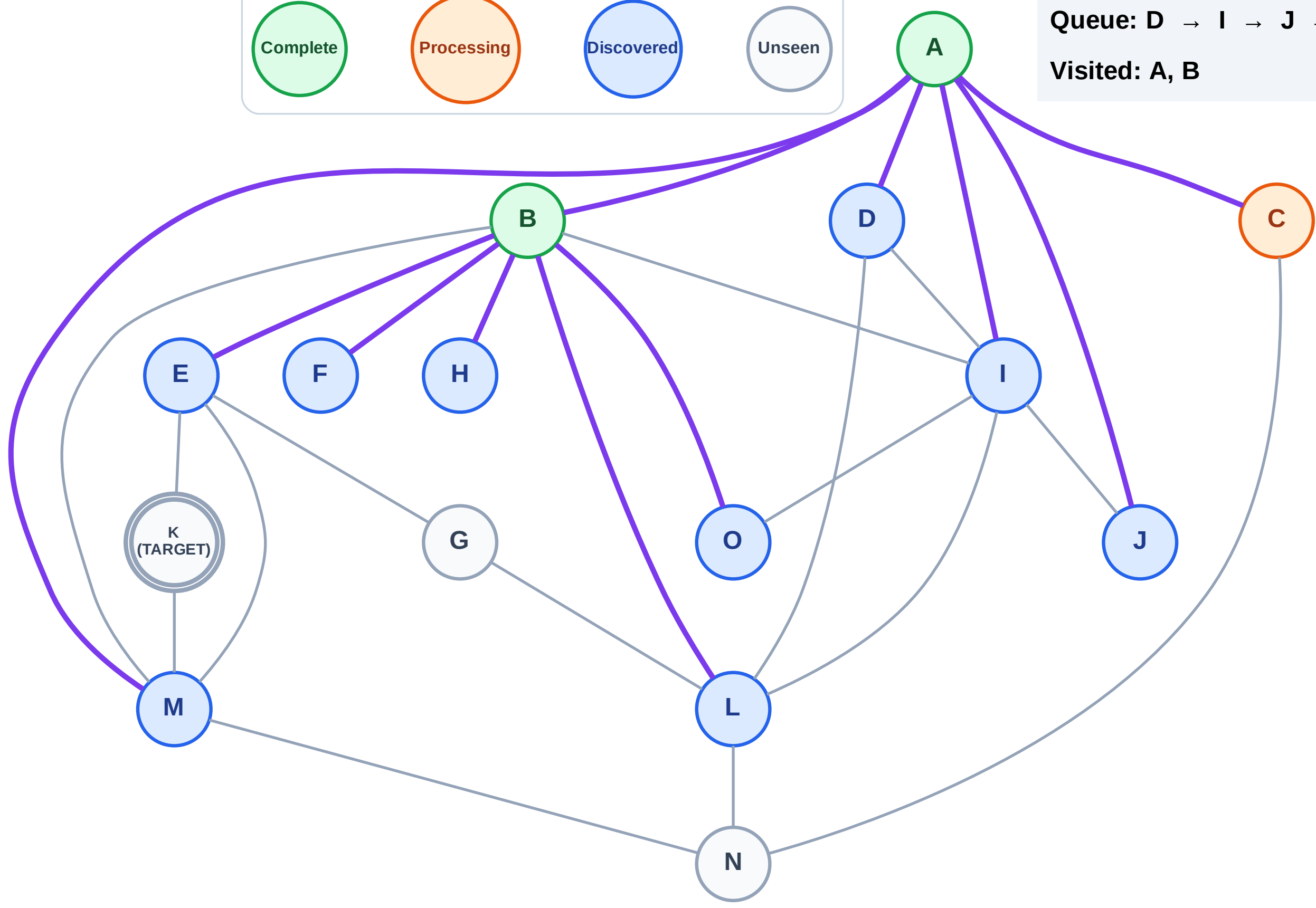
BFS Traversal

Step 6: Process node C

Legend



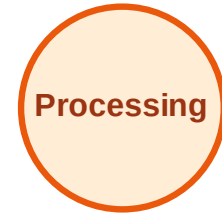
Queue: D → I → J → M → E → F → H → L → O
Visited: A, B



BFS Traversal

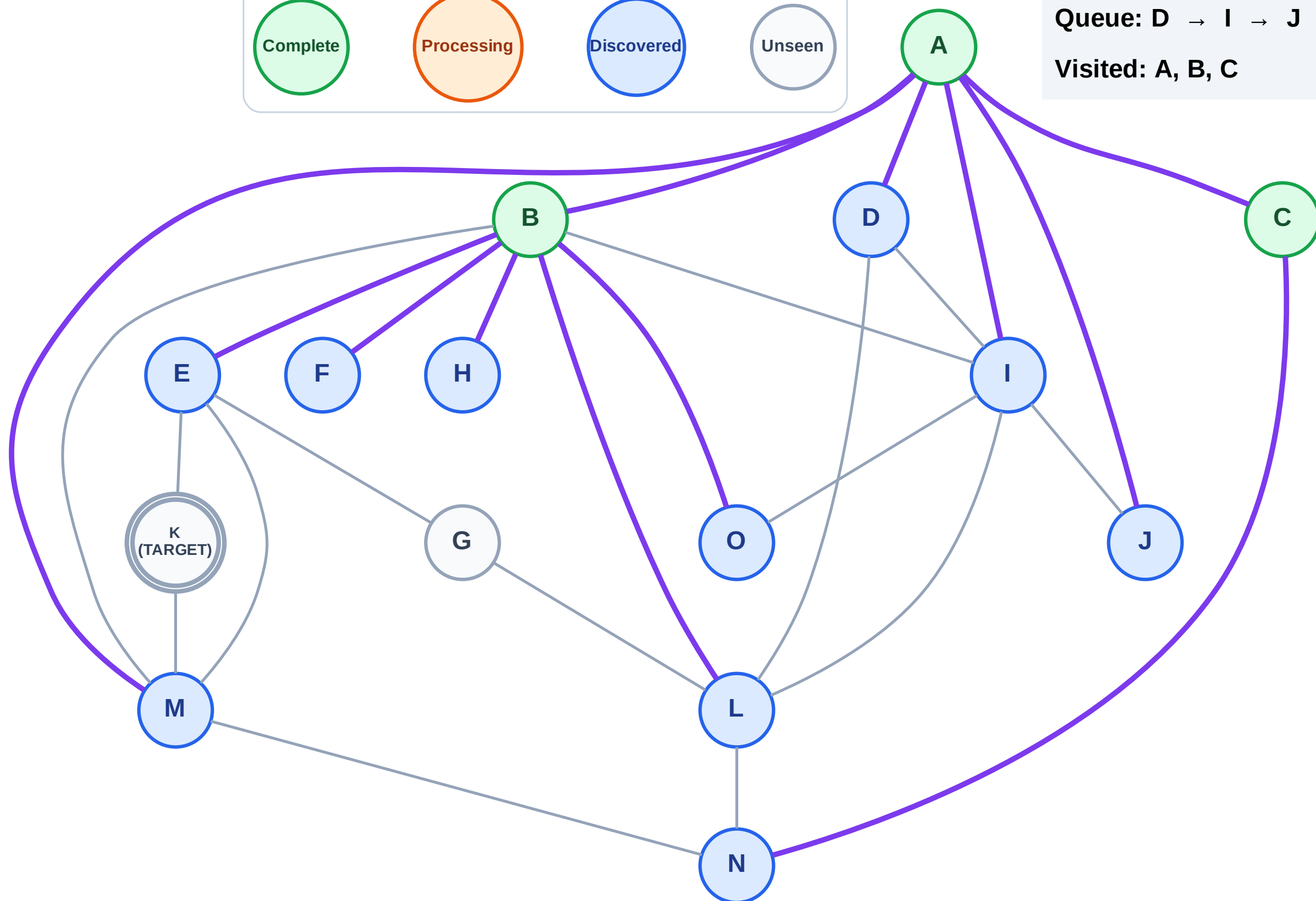
Step 7: Discover N from C

Legend



Queue: D → I → J → M → E → F → H → L → O → N

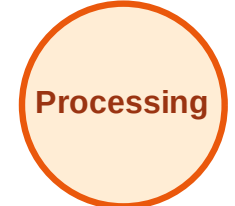
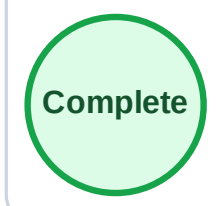
Visited: A, B, C



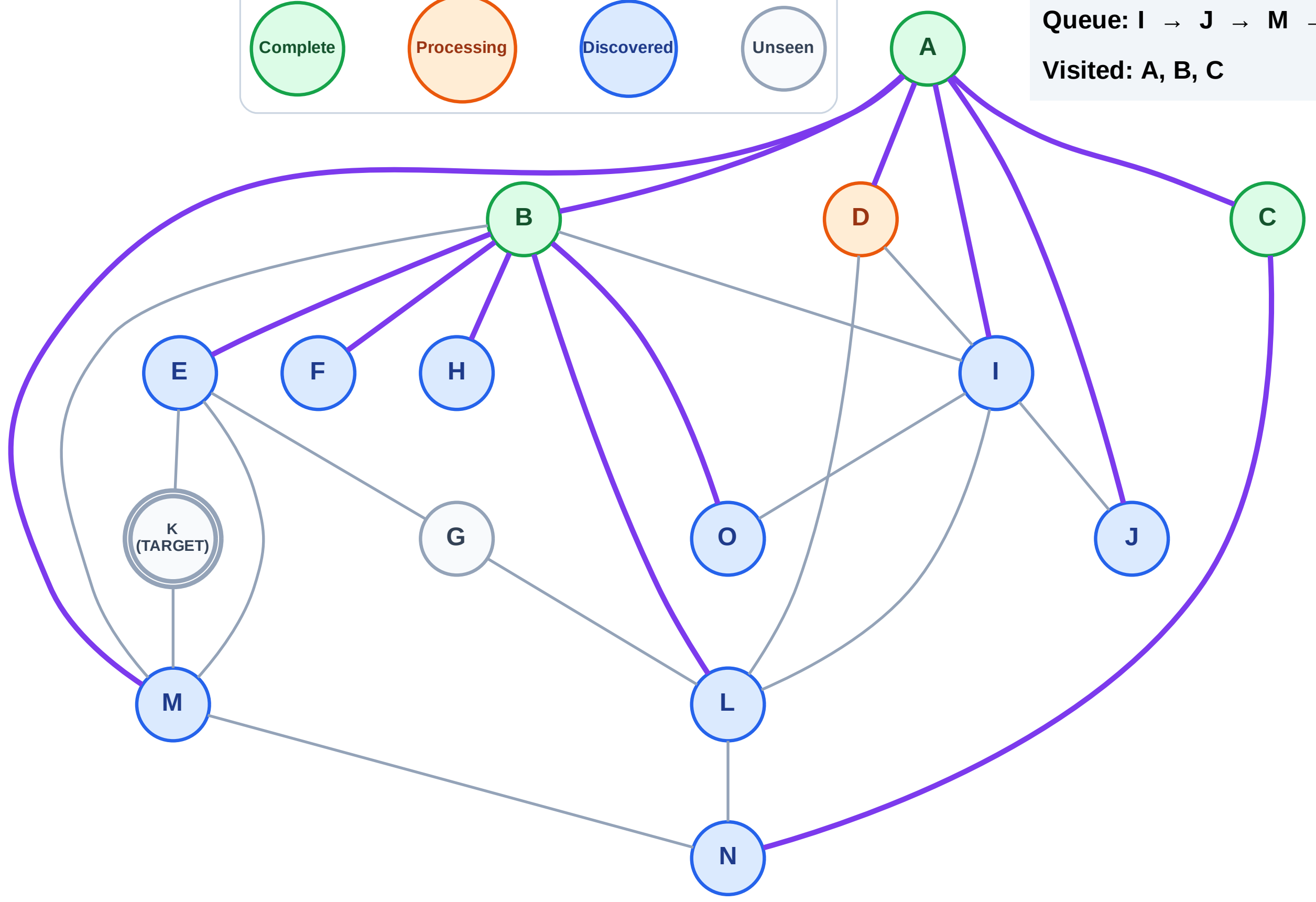
BFS Traversal

Step 8: Process node D

Legend



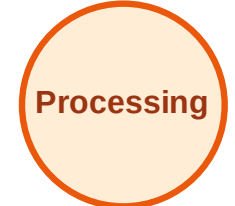
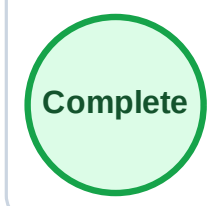
Queue: I → J → M → E → F → H → L → O → N
Visited: A, B, C



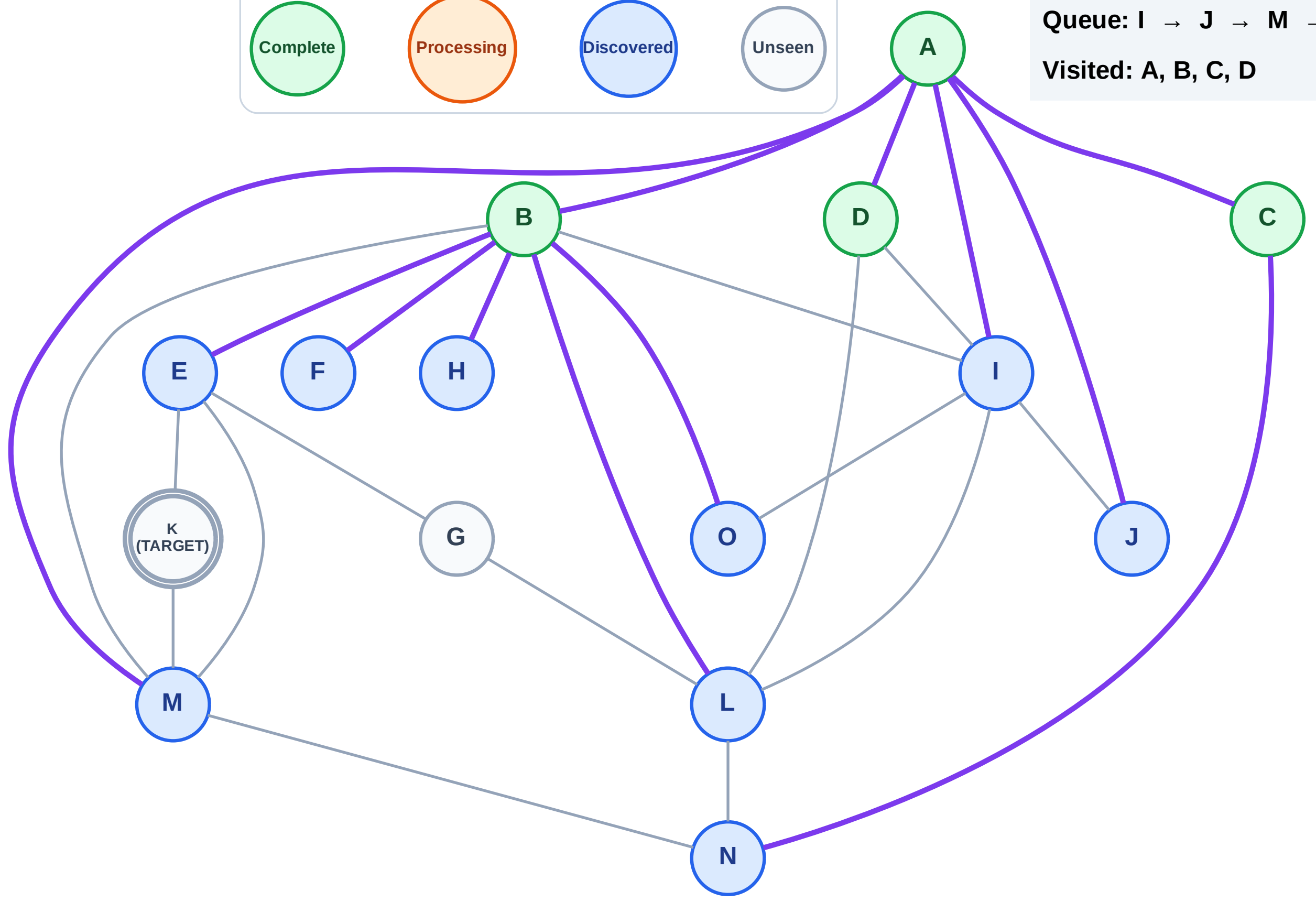
BFS Traversal

Step 9: No new neighbors from D

Legend



Queue: I → J → M → E → F → H → L → O → N
Visited: A, B, C, D



BFS Traversal

Step 10: Process node I

Legend

Complete

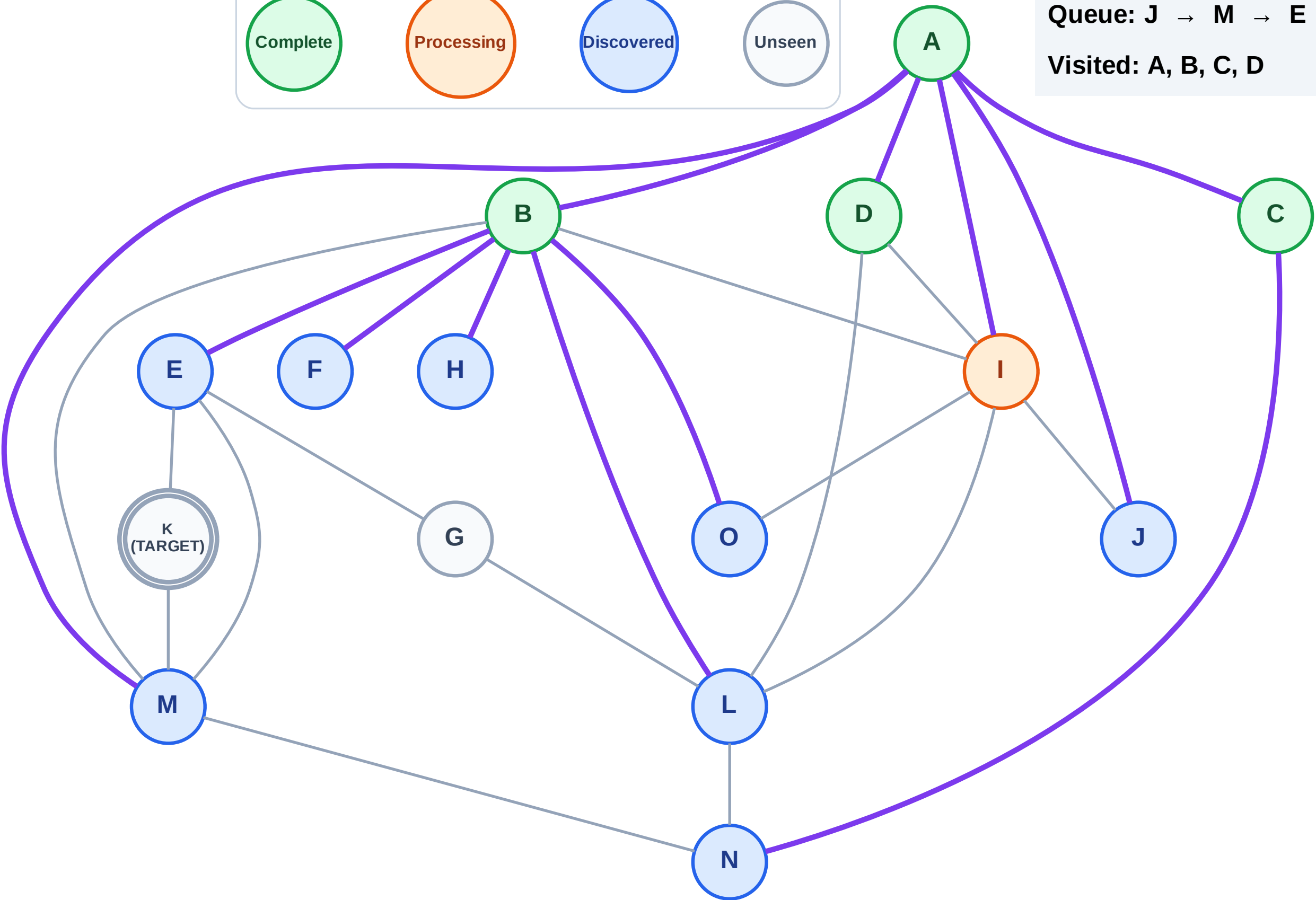
Processing

Discovered

Unseen

Queue: J → M → E → F → H → L → O → N

Visited: A, B, C, D



BFS Traversal

Step 11: No new neighbors from I

Legend

Complete

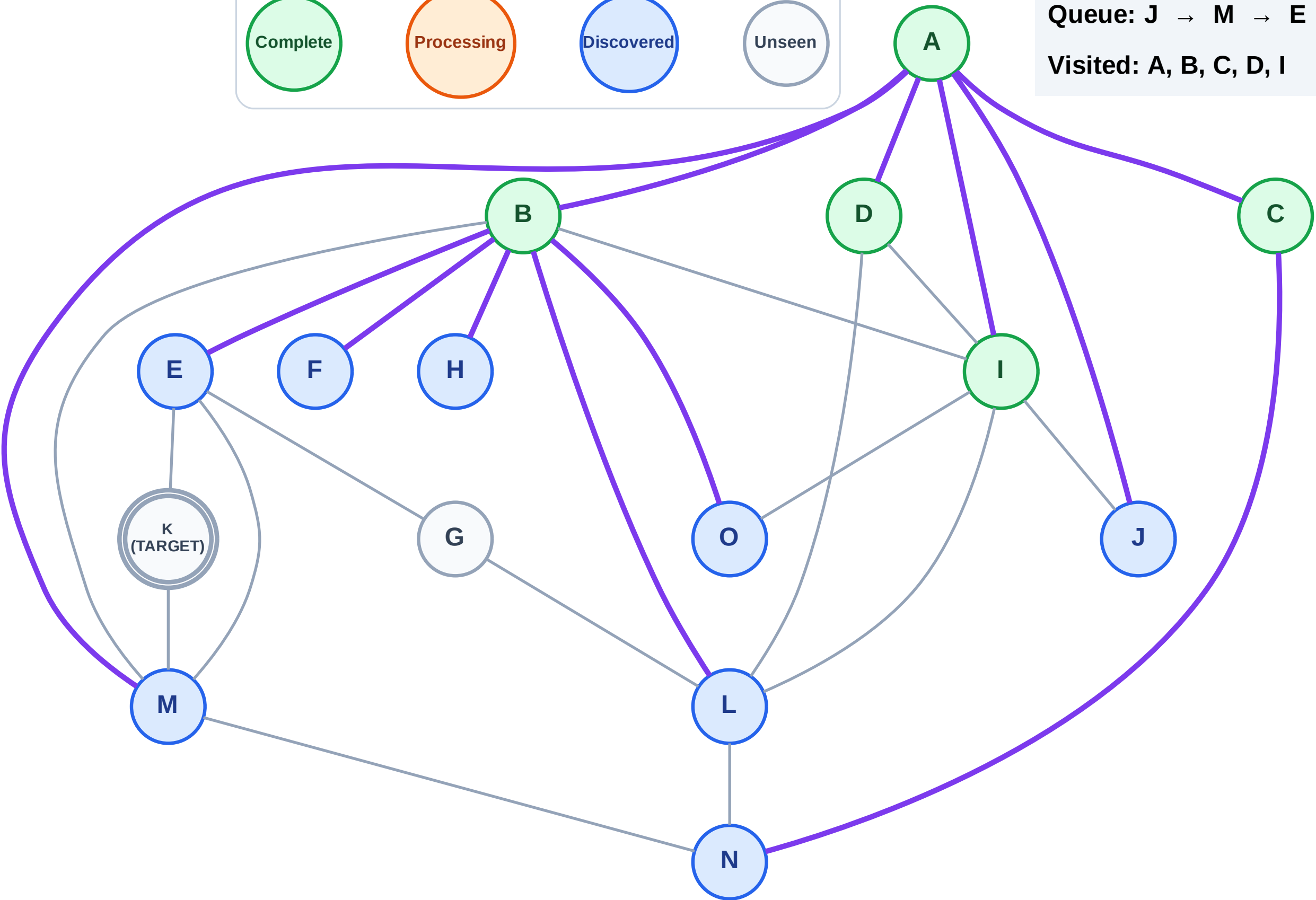
Processing

Discovered

Unseen

Queue: J → M → E → F → H → L → O → N

Visited: A, B, C, D, I



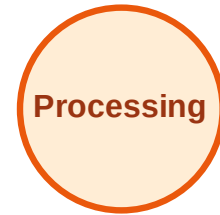
BFS Traversal

Step 12: Process node J

Legend



Complete



Processing

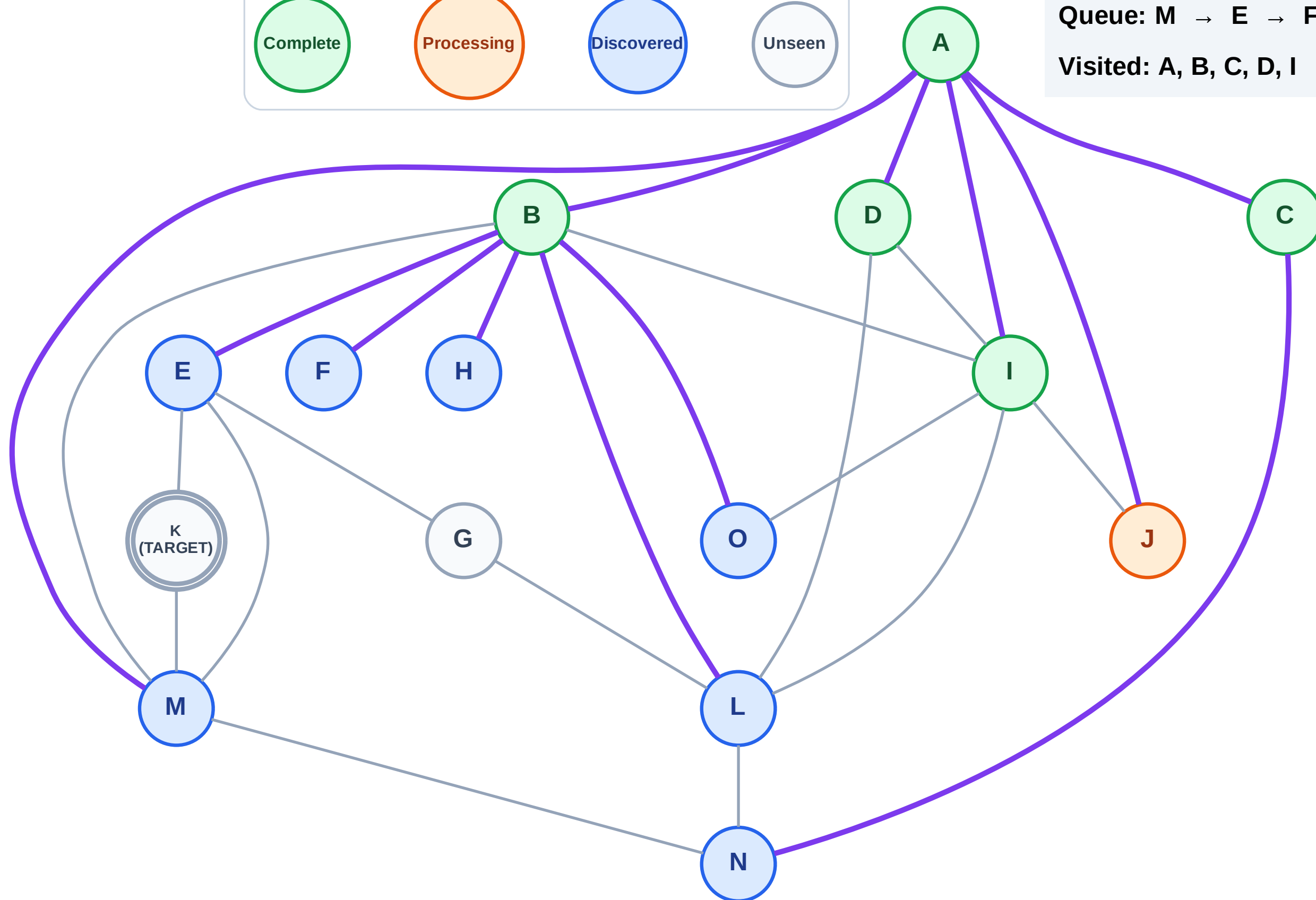


Discovered



Unseen

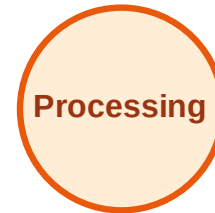
Queue: M → E → F → H → L → O → N
Visited: A, B, C, D, I



BFS Traversal

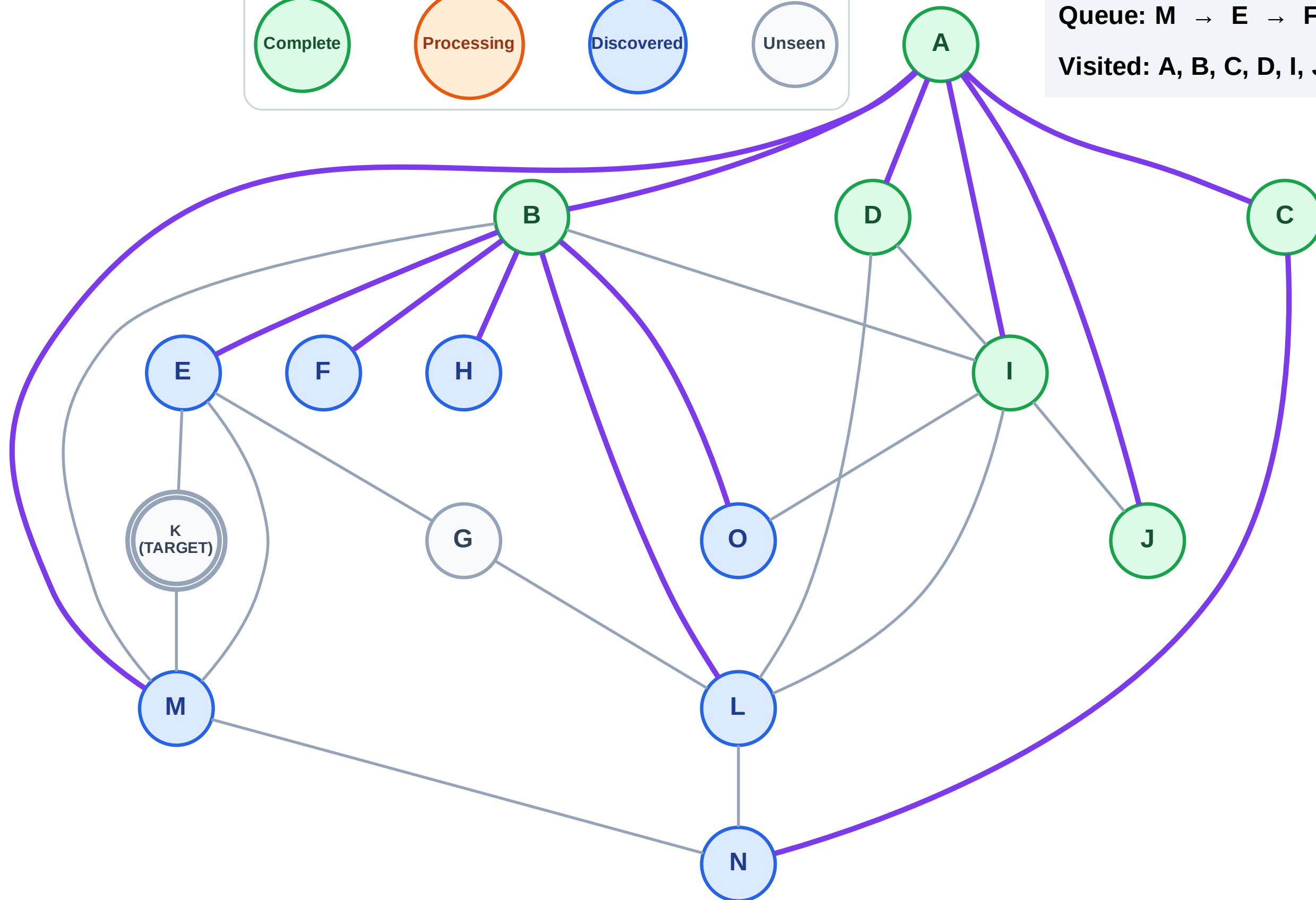
Step 13: No new neighbors from J

Legend



Queue: M → E → F → H → L → O → N

Visited: A, B, C, D, I, J



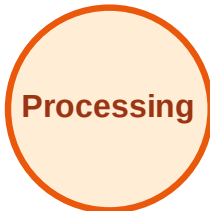
BFS Traversal

Step 14: Process node M

Legend



Complete



Processing



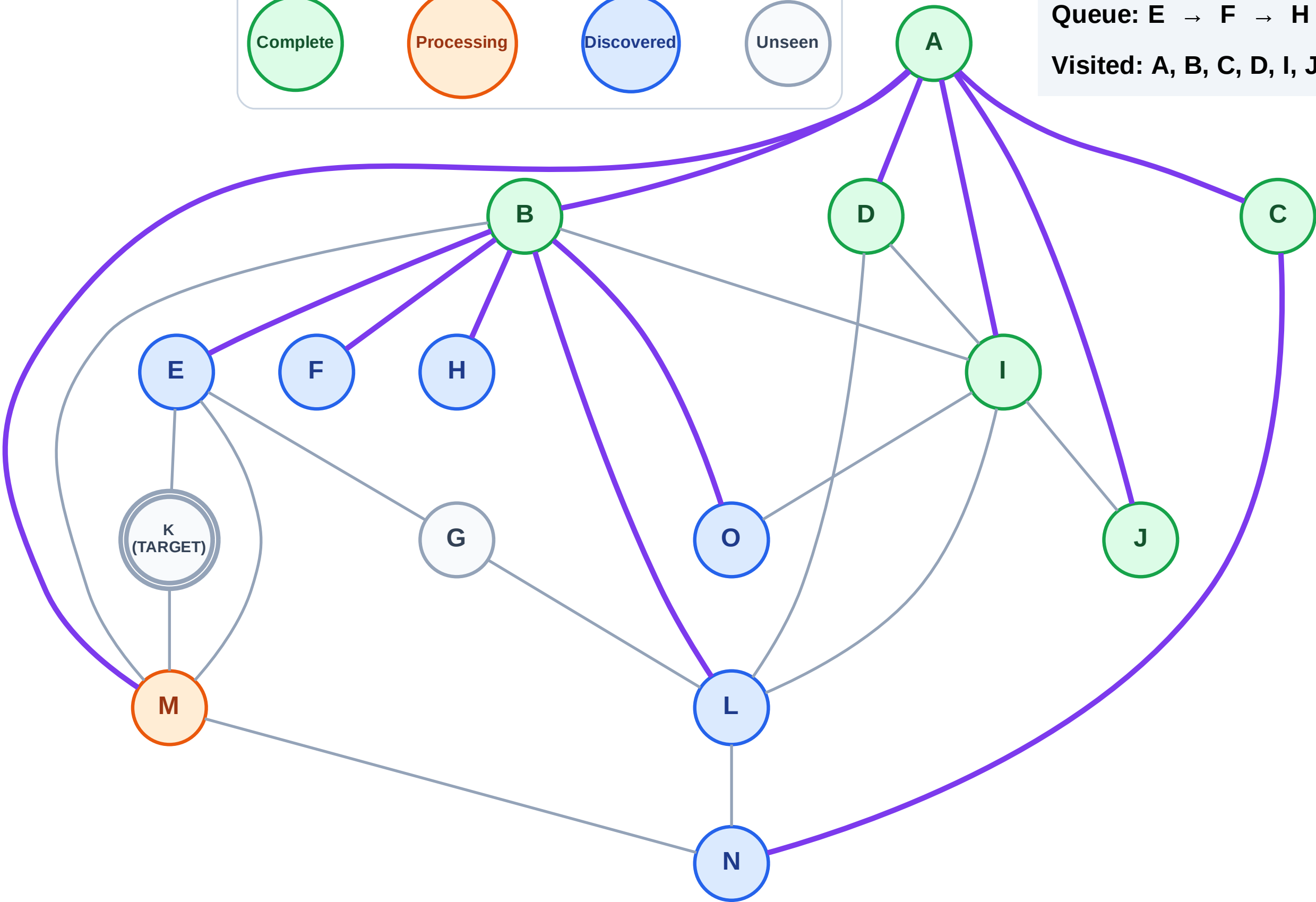
Discovered



Unseen

Queue: E → F → H → L → O → N

Visited: A, B, C, D, I, J



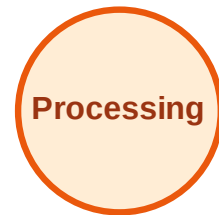
BFS Traversal

Step 15: Discover K from M

Legend



Complete



Processing



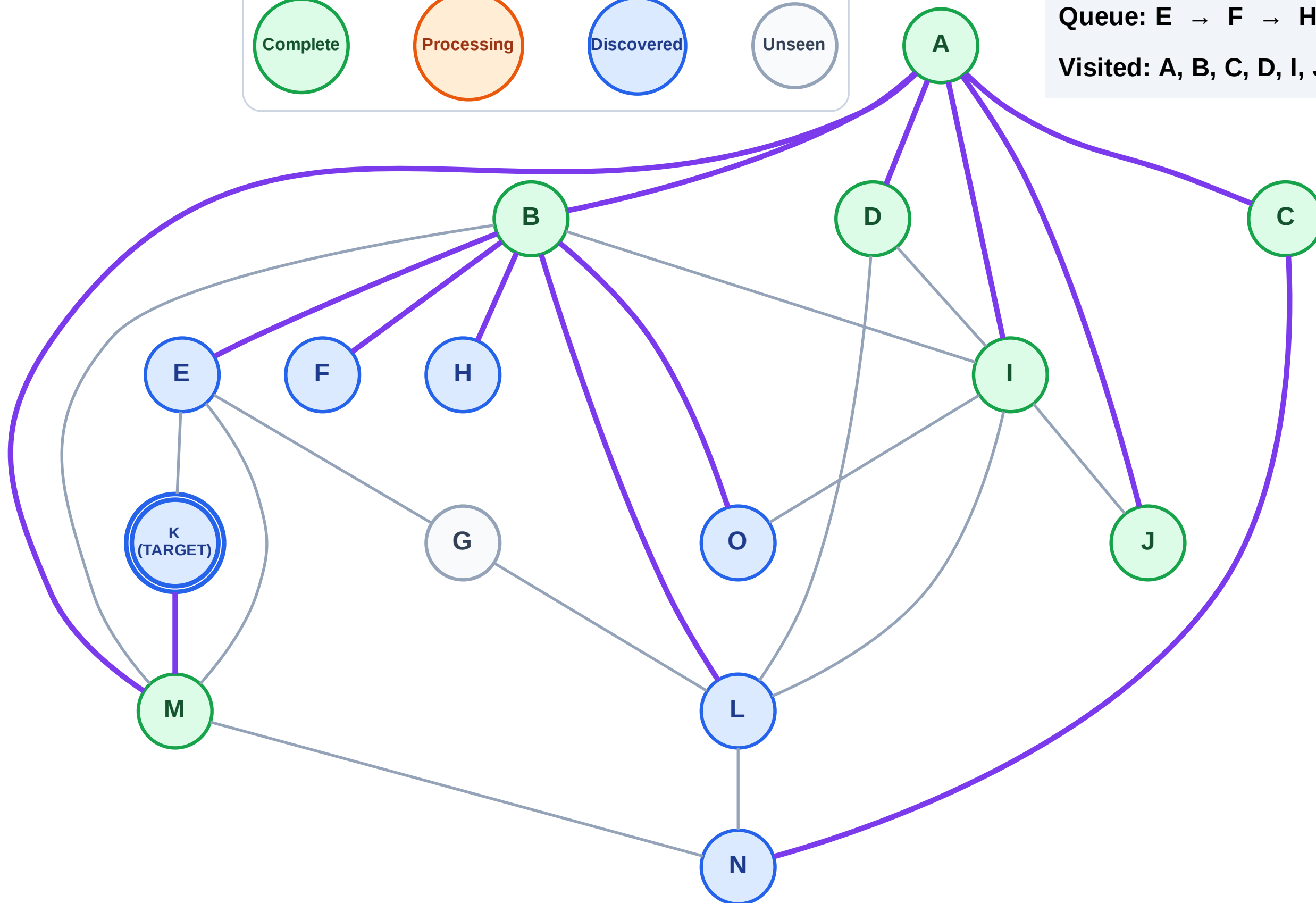
Discovered



Unseen

Queue: E → F → H → L → O → N → K

Visited: A, B, C, D, I, J, M



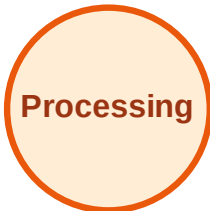
BFS Traversal

Step 16: Process node E

Legend



Complete



Processing



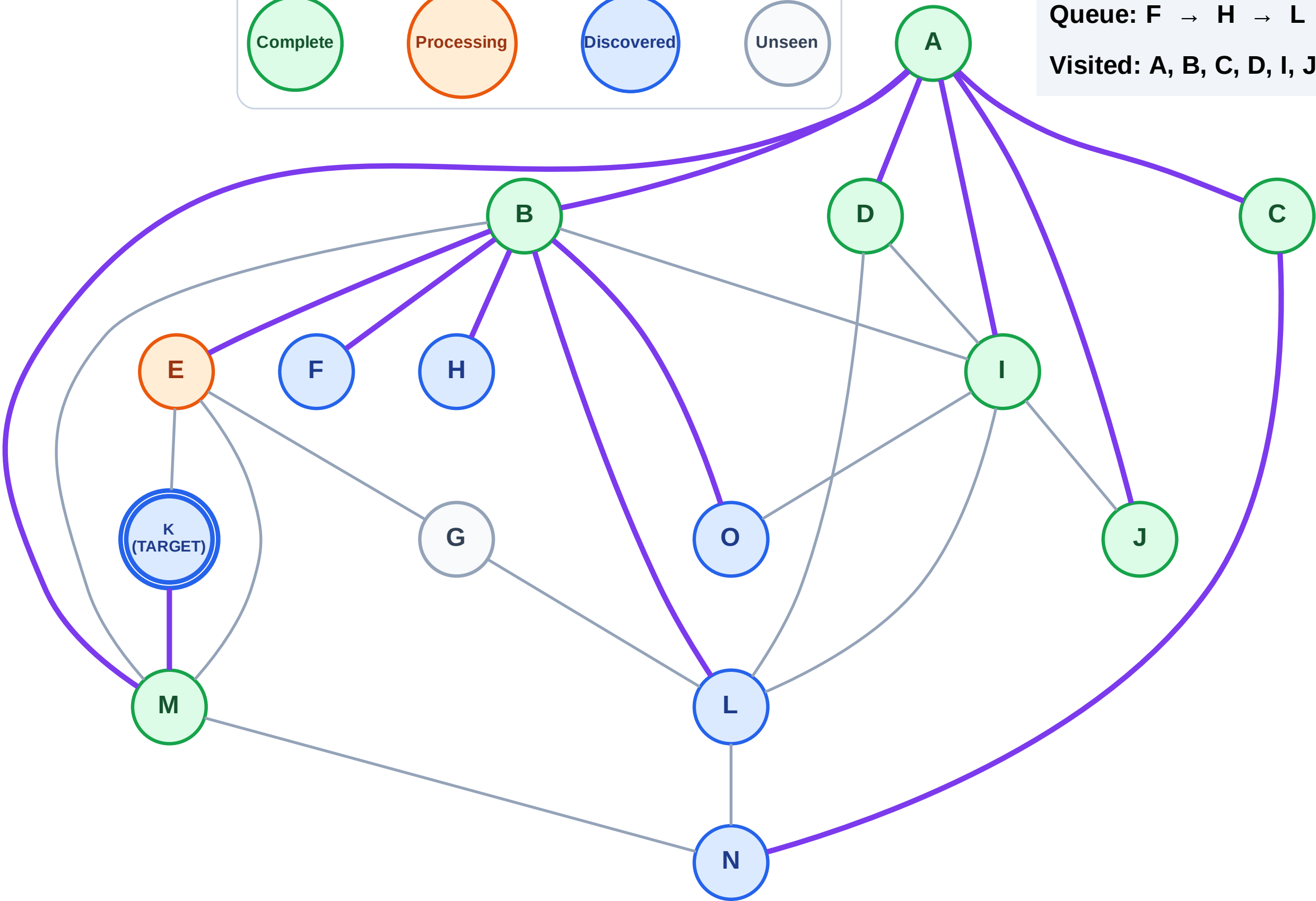
Discovered



Unseen

Queue: F → H → L → O → N → K

Visited: A, B, C, D, I, J, M



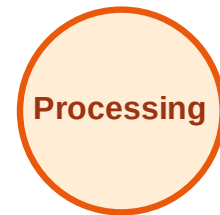
BFS Traversal

Step 17: Discover G from E

Legend



Complete



Processing



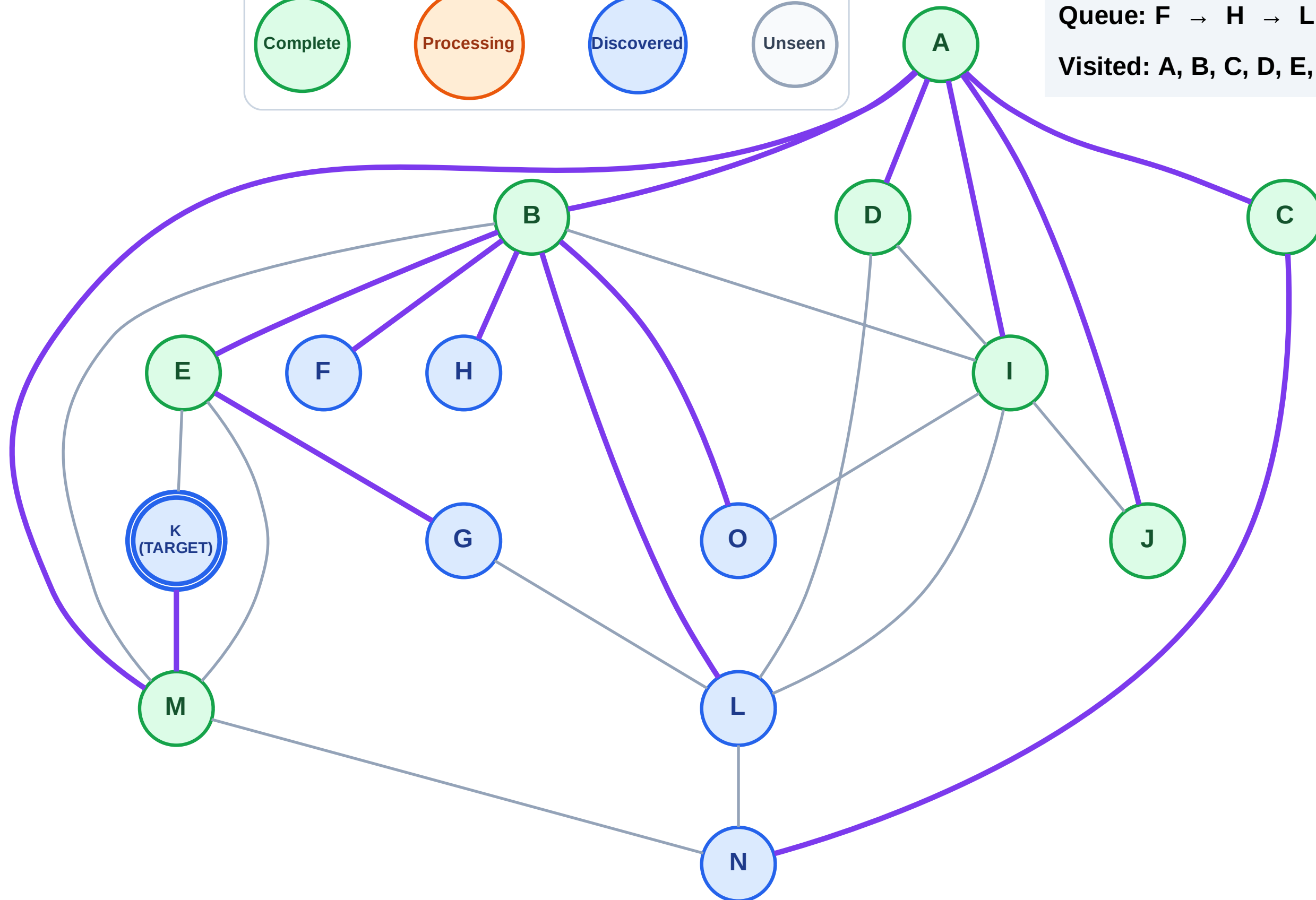
Discovered



Unseen

Queue: F → H → L → O → N → K → G

Visited: A, B, C, D, E, I, J, M



BFS Traversal

Step 18: Process node F

Legend

Complete

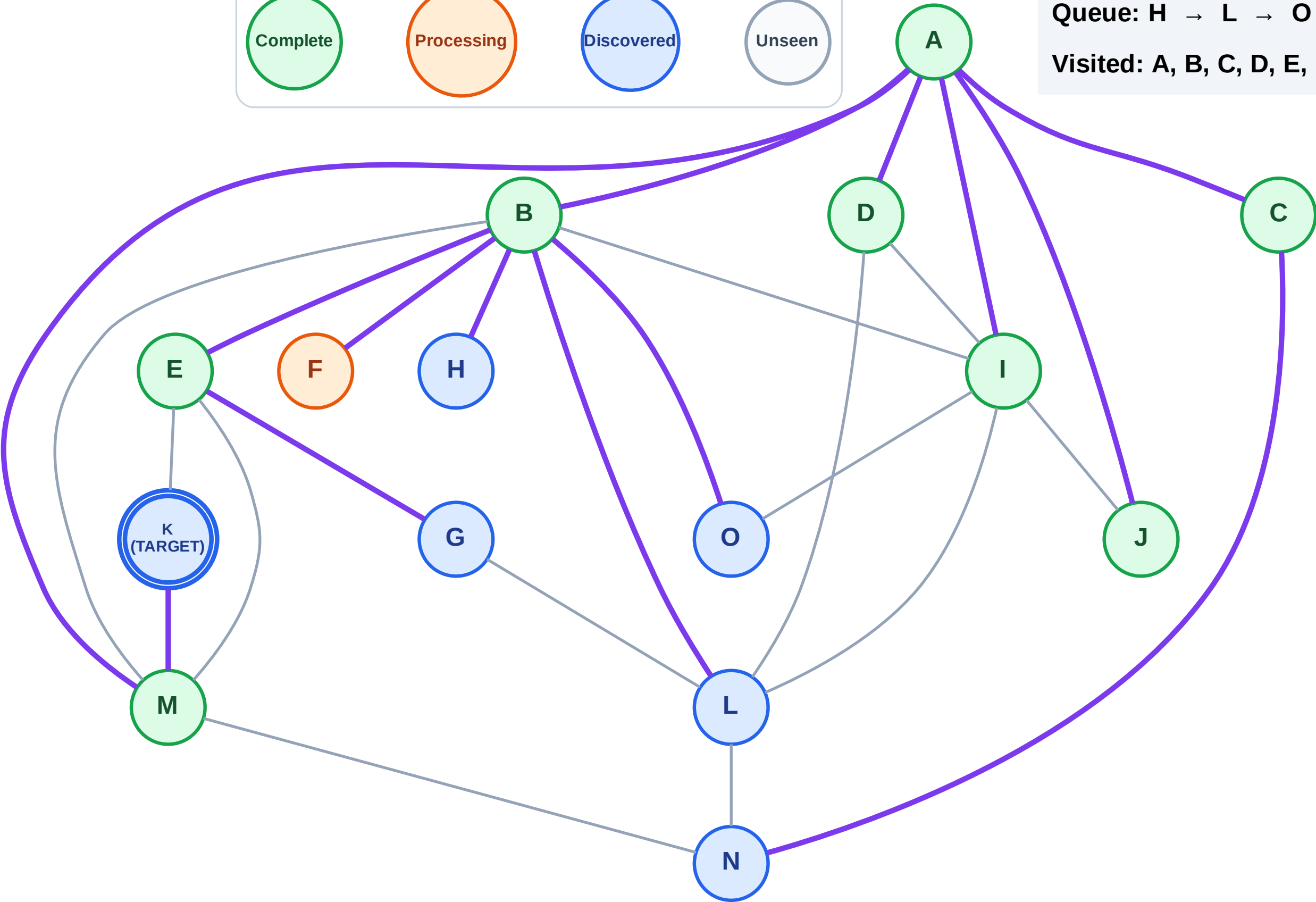
Processing

Discovered

Unseen

Queue: H → L → O → N → K → G

Visited: A, B, C, D, E, I, J, M



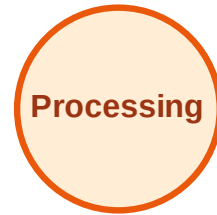
BFS Traversal

Step 19: No new neighbors from F

Legend



Complete



Processing



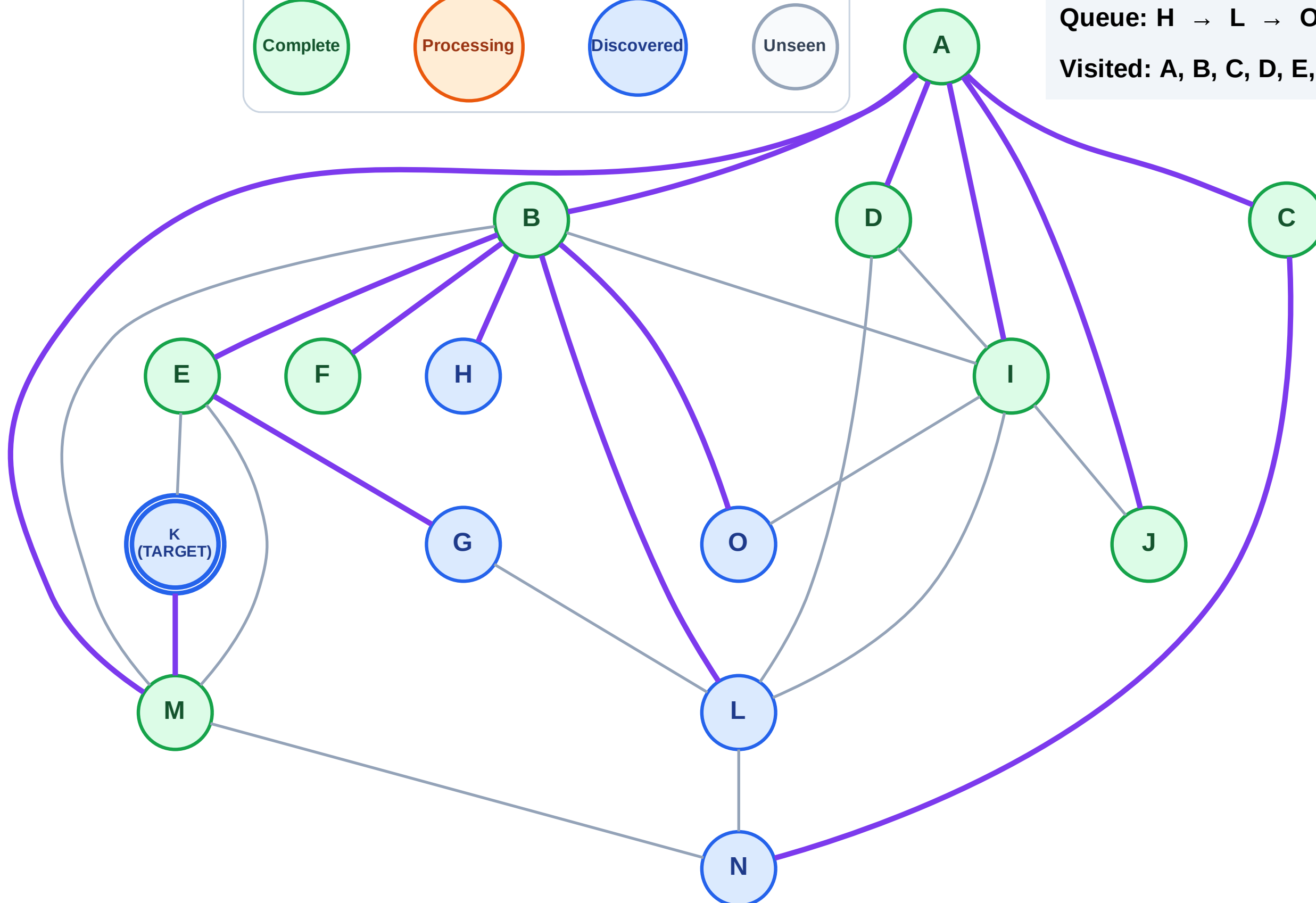
Discovered



Unseen

Queue: H → L → O → N → K → G

Visited: A, B, C, D, E, F, I, J, M



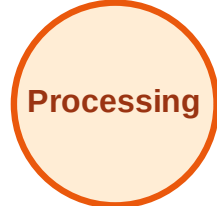
BFS Traversal

Step 20: Process node H

Legend



Complete



Processing



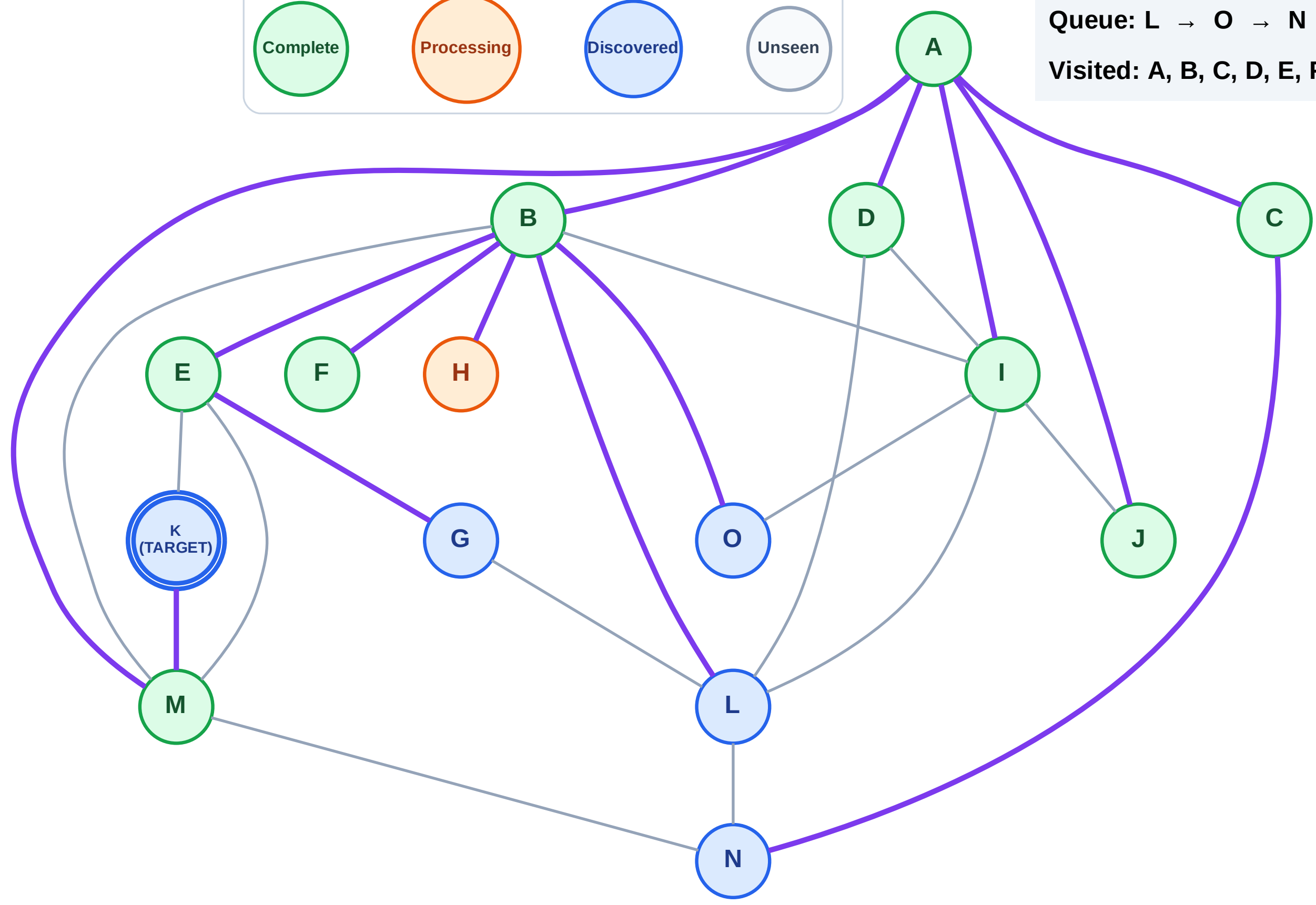
Discovered



Unseen

Queue: L → O → N → K → G

Visited: A, B, C, D, E, F, I, J, M



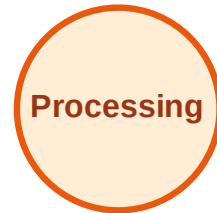
BFS Traversal

Step 21: No new neighbors from H

Legend



Complete



Processing



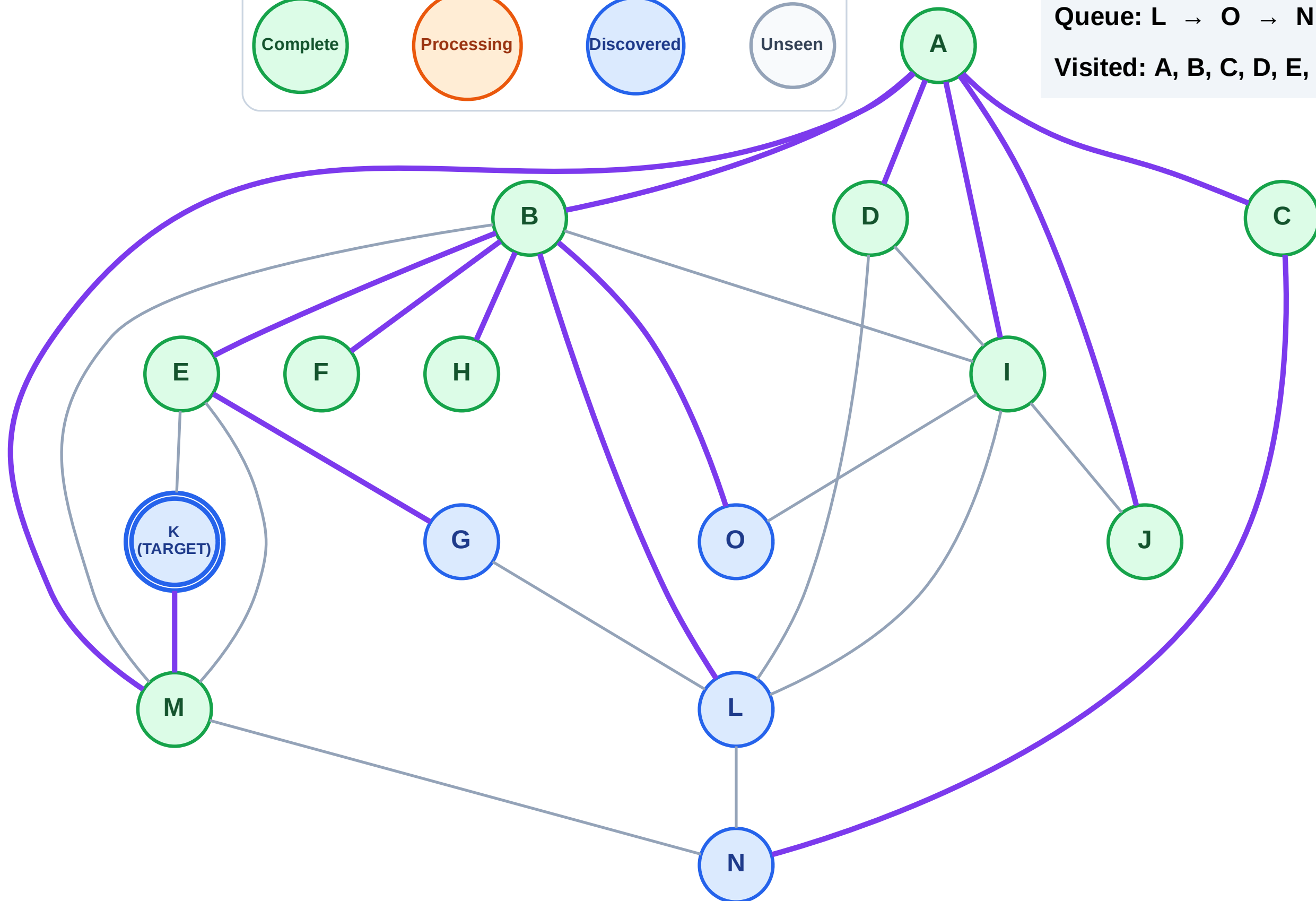
Discovered



Unseen

Queue: L → O → N → K → G

Visited: A, B, C, D, E, F, H, I, J, M



BFS Traversal

Step 22: Process node L

Legend



Complete



Processing



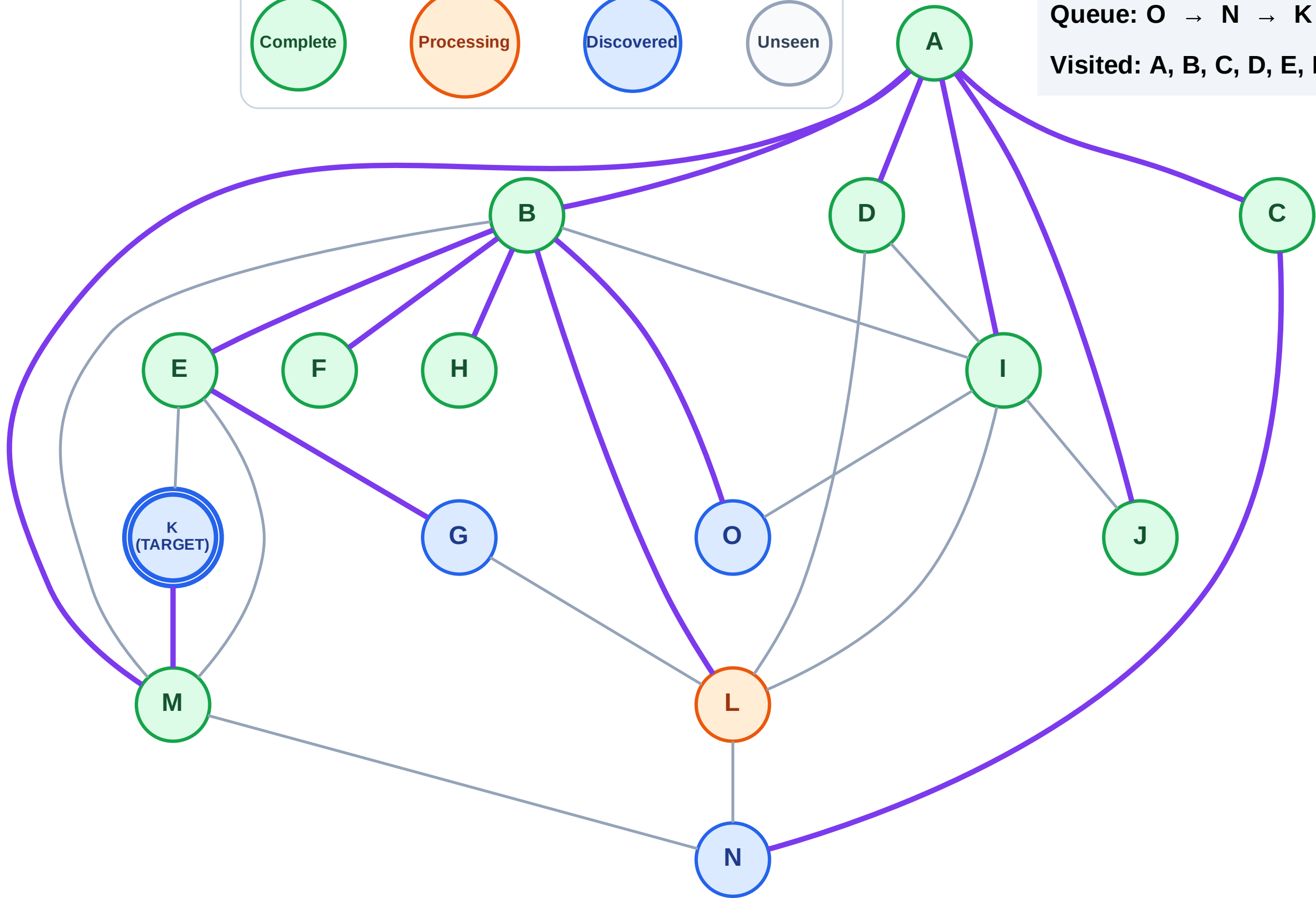
Discovered



Unseen

Queue: O → N → K → G

Visited: A, B, C, D, E, F, H, I, J, M



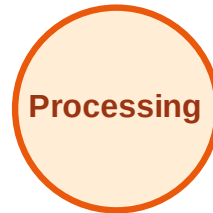
BFS Traversal

Step 23: No new neighbors from L

Legend



Complete



Processing



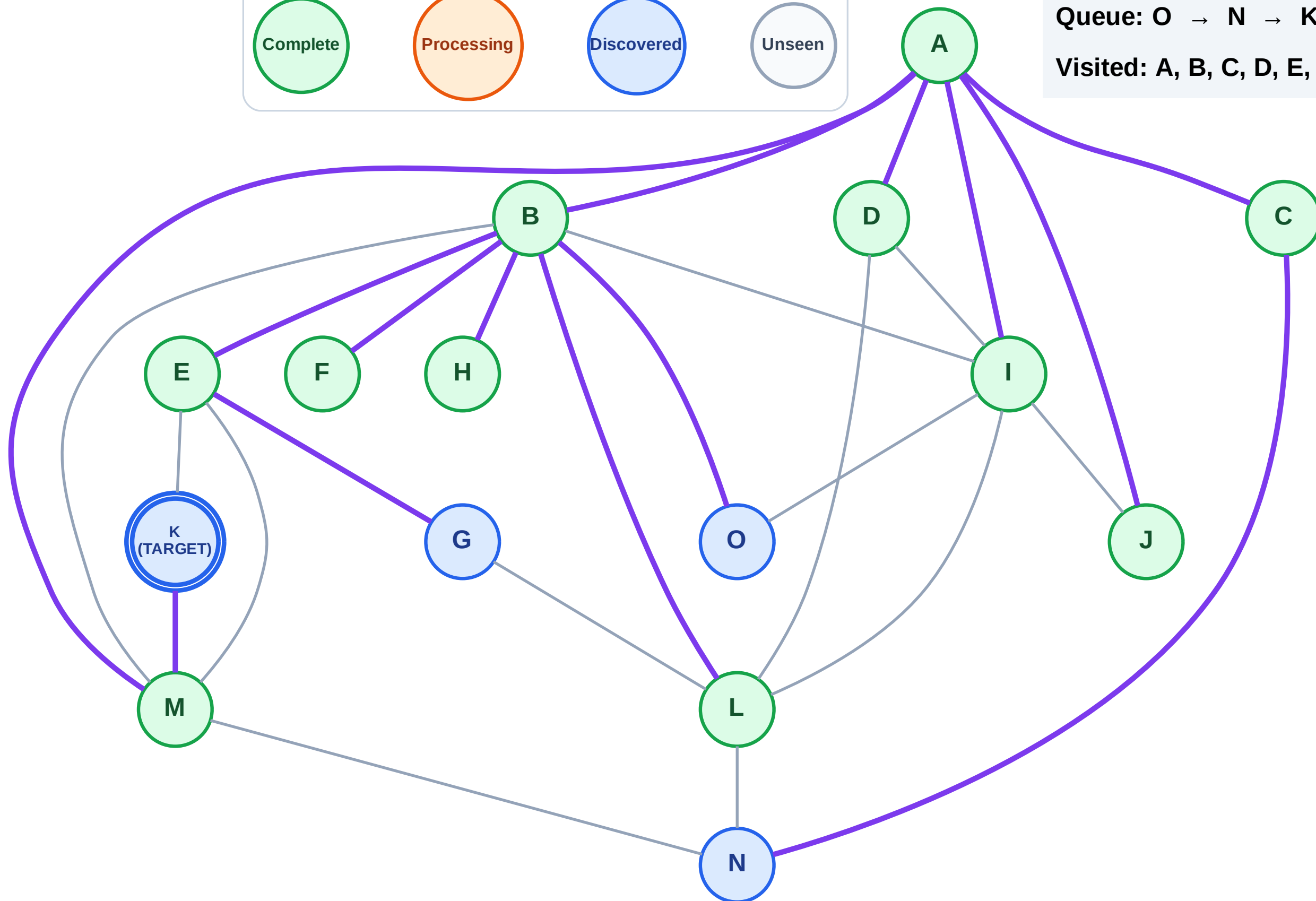
Discovered



Unseen

Queue: O → N → K → G

Visited: A, B, C, D, E, F, H, I, J, L, M



BFS Traversal

Step 24: Process node O

Legend

Complete

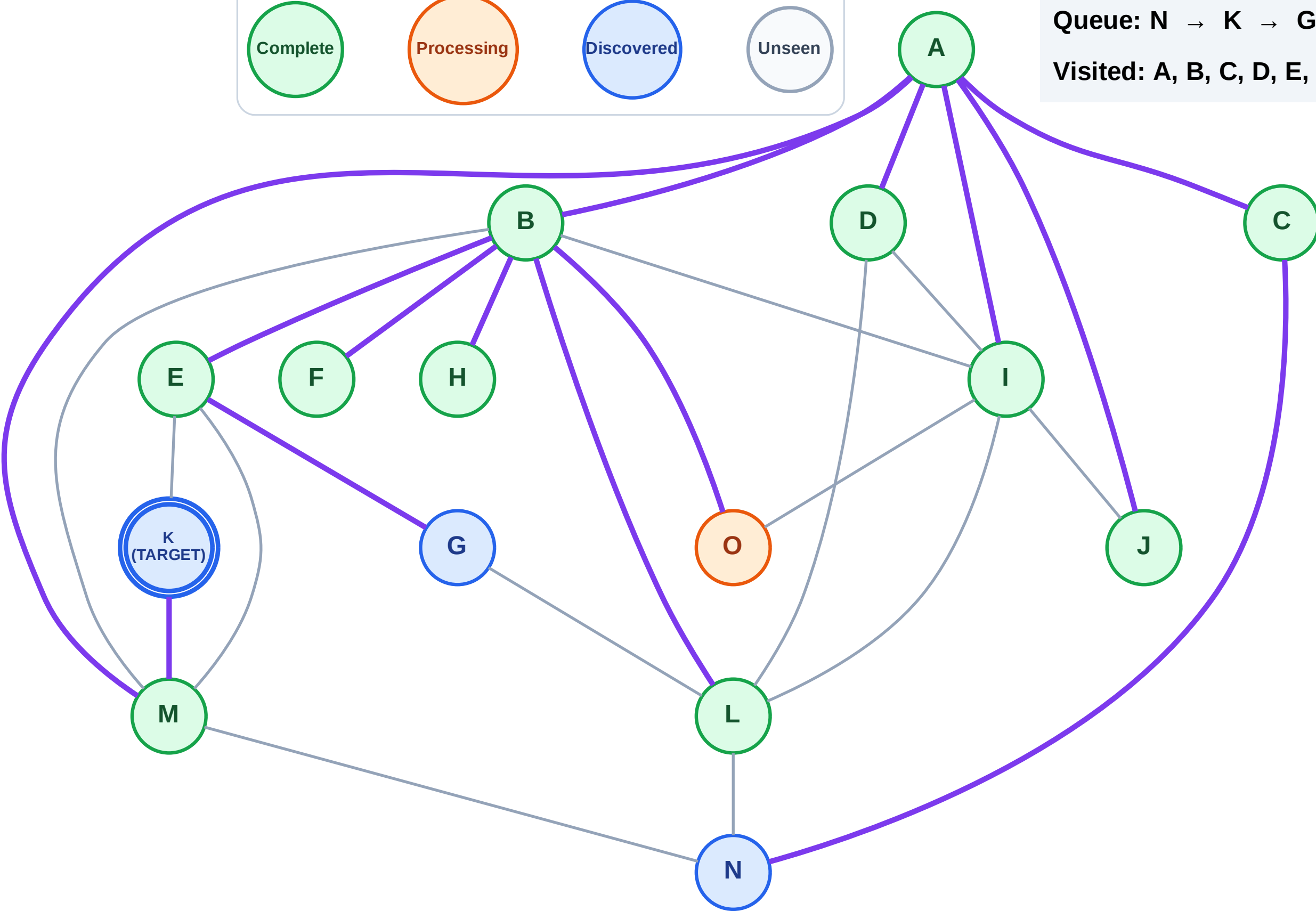
Processing

Discovered

Unseen

Queue: N → K → G

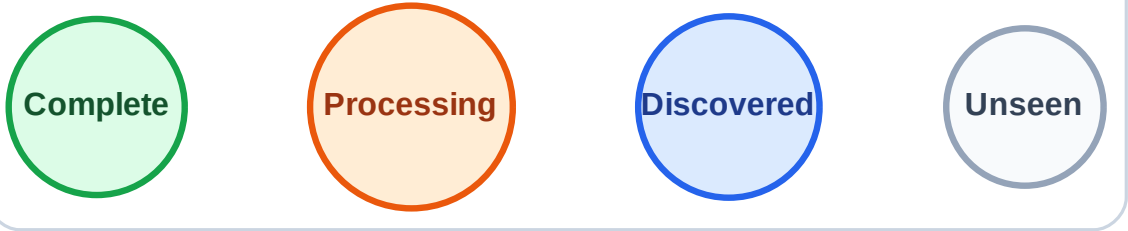
Visited: A, B, C, D, E, F, H, I, J, L, M



BFS Traversal

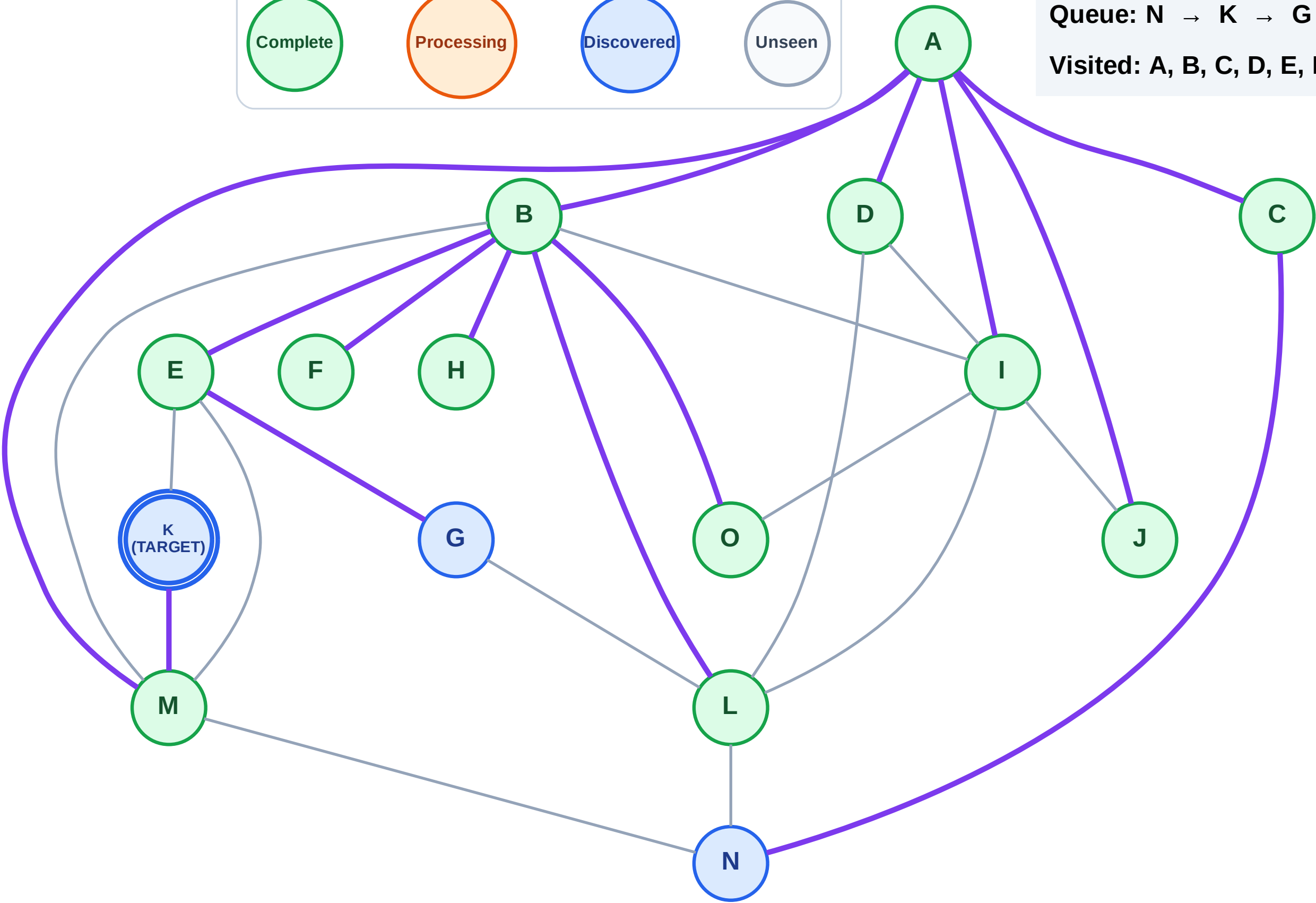
Step 25: No new neighbors from O

Legend



Queue: N → K → G

Visited: A, B, C, D, E, F, H, I, J, L, M, O



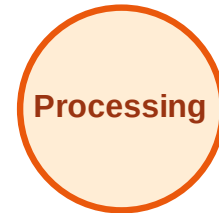
BFS Traversal

Step 26: Process node N

Legend



Complete



Processing



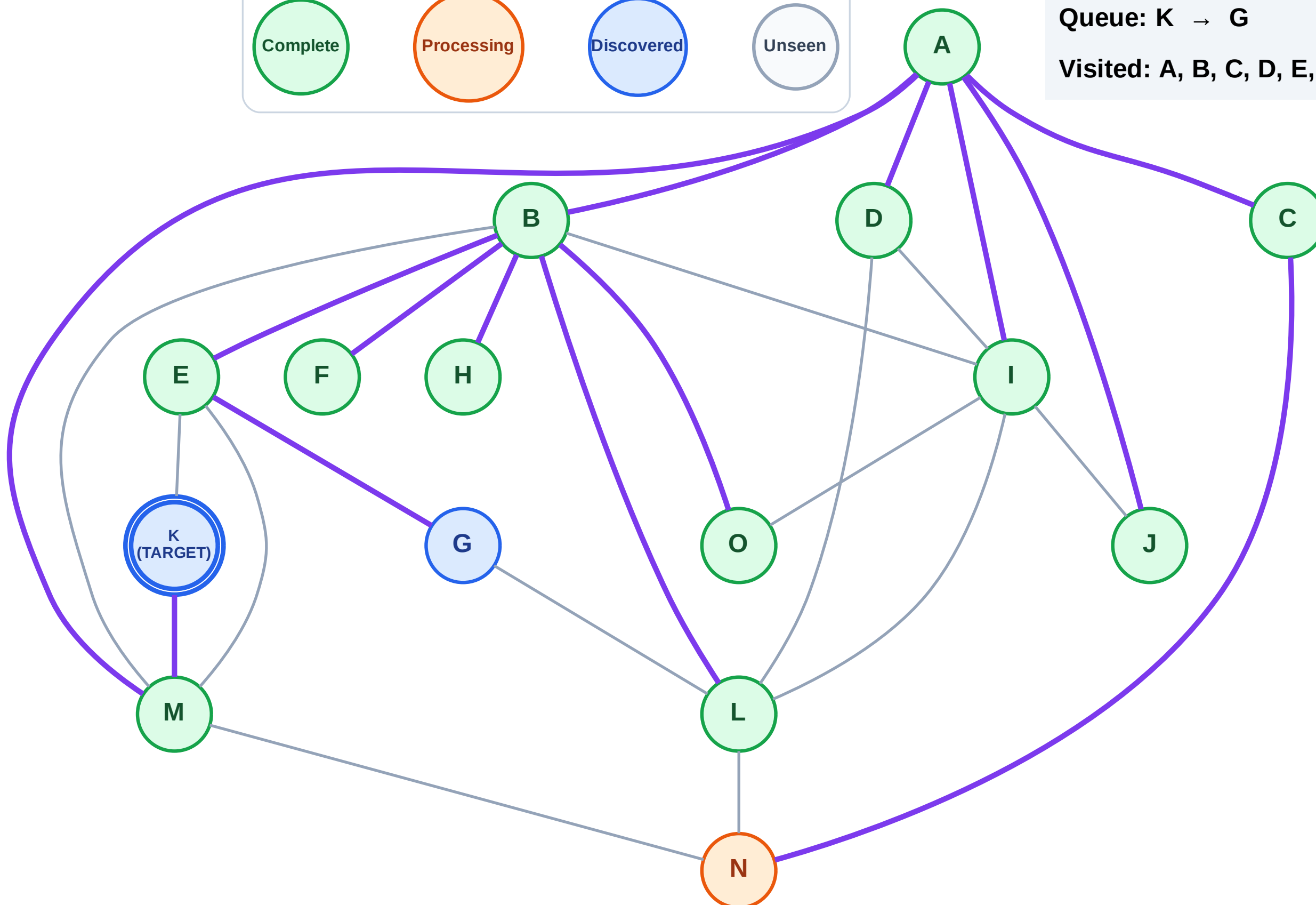
Discovered



Unseen

Queue: K → G

Visited: A, B, C, D, E, F, H, I, J, L, M, O



BFS Traversal

Step 27: No new neighbors from N

Legend



Complete



Processing



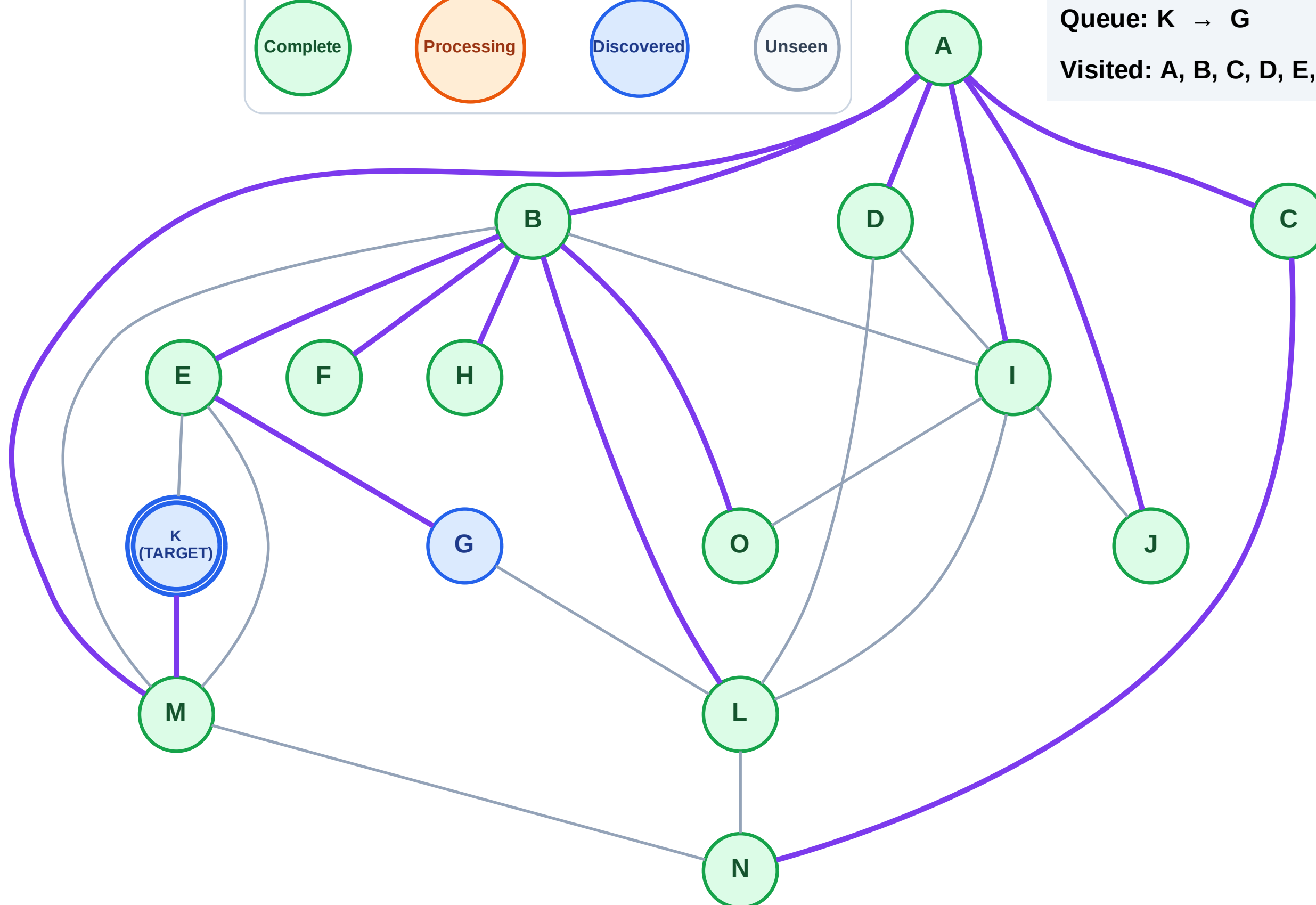
Discovered



Unseen

Queue: K → G

Visited: A, B, C, D, E, F, H, I, J, L, M, N, O



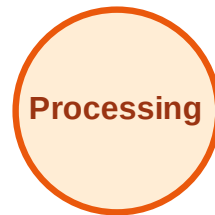
BFS Traversal

Step 28: Process node K

Legend



Complete



Processing



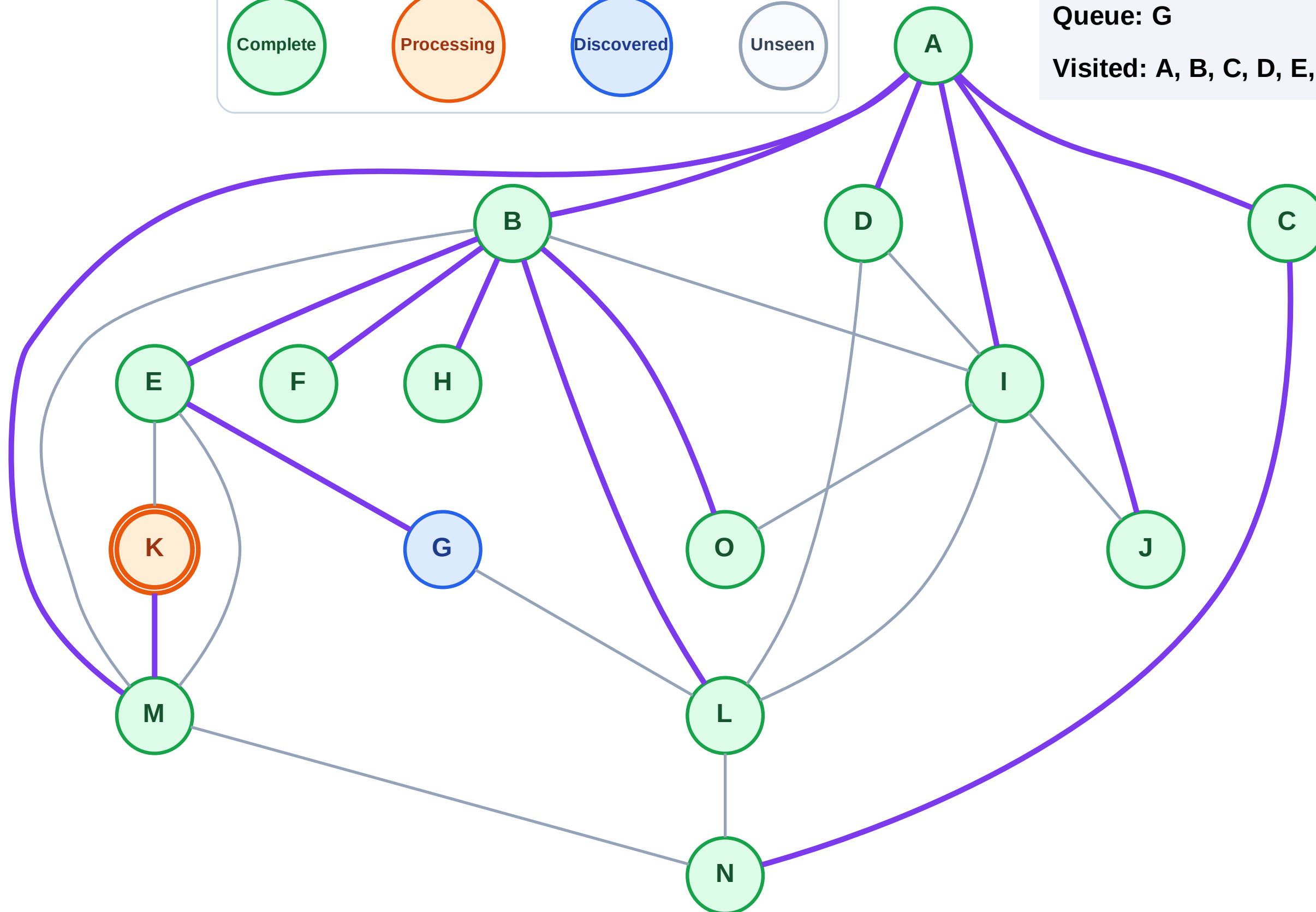
Discovered



Unseen

Queue: G

Visited: A, B, C, D, E, F, H, I, J, L, M, N, O



BFS Traversal

Step 29: TARGET NODE K REACHED! Path (2 hops): A → M → K

Legend

Complete

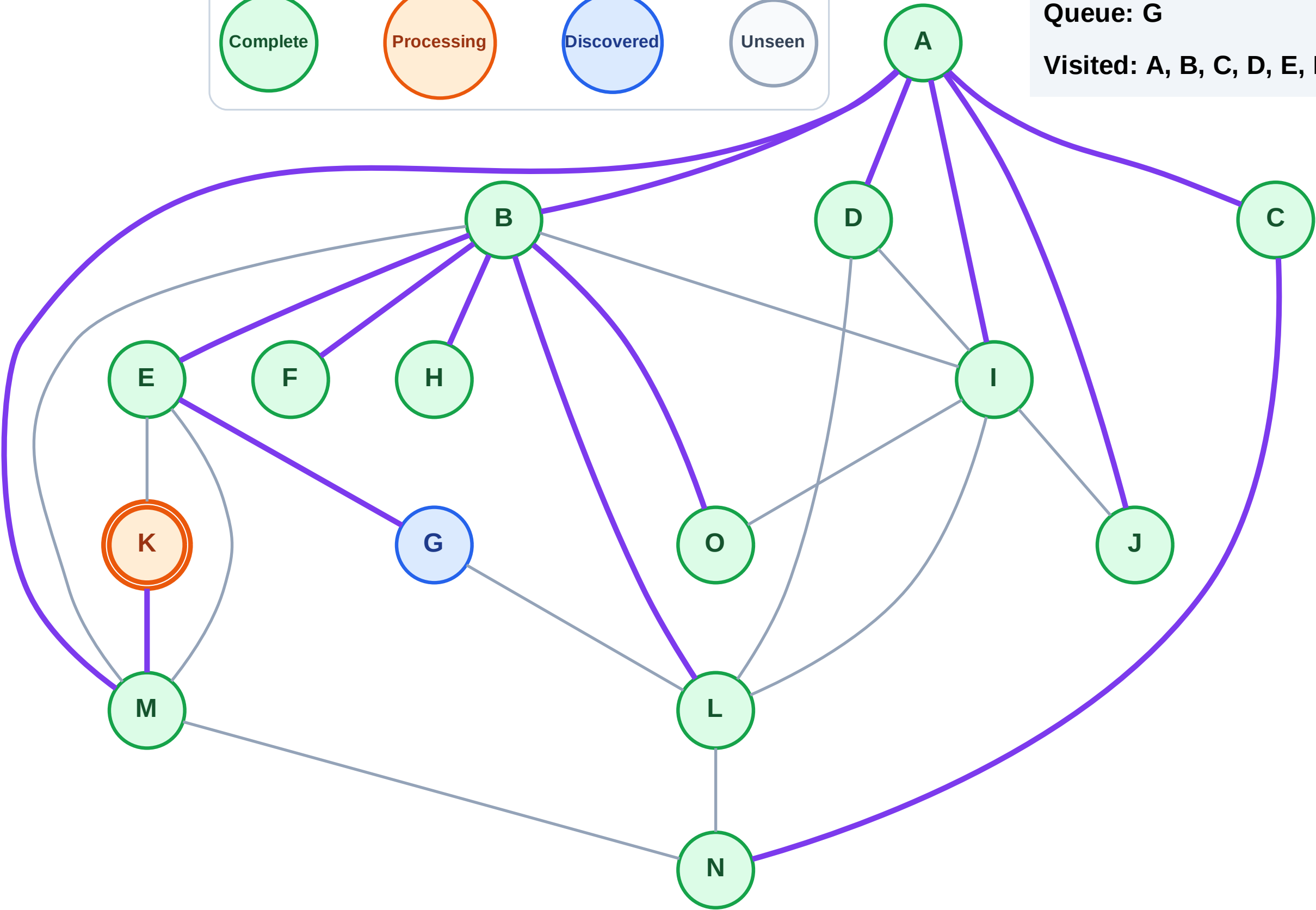
Processing

Discovered

Unseen

Queue: G

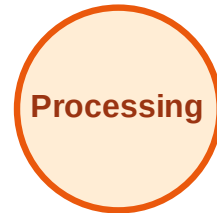
Visited: A, B, C, D, E, F, H, I, J, K, L, M, N, O



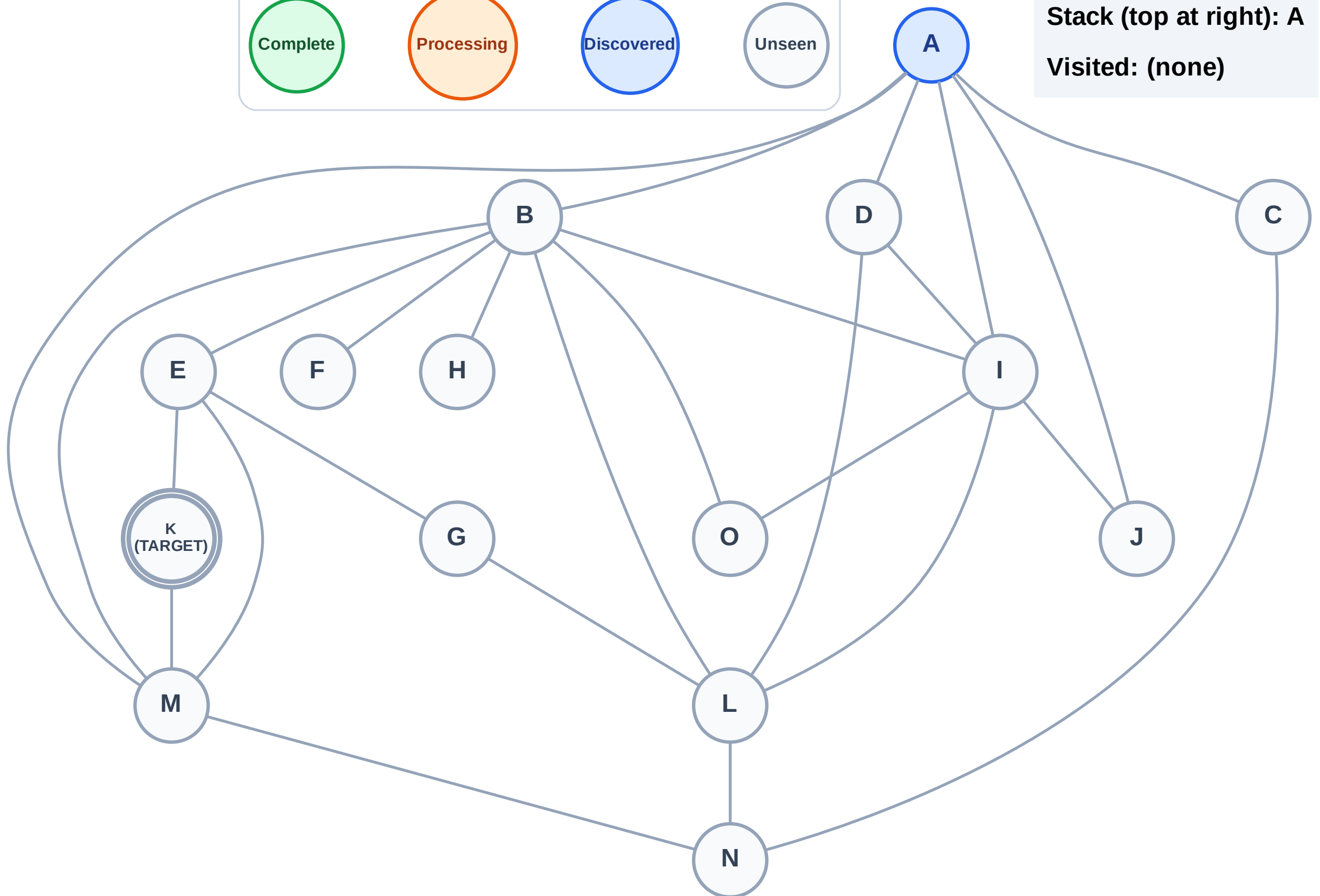
DFS Traversal

Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A
Visited: (none)



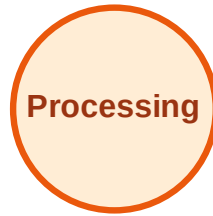
DFS Traversal

Step 2: Process node A

Legend



Complete



Processing



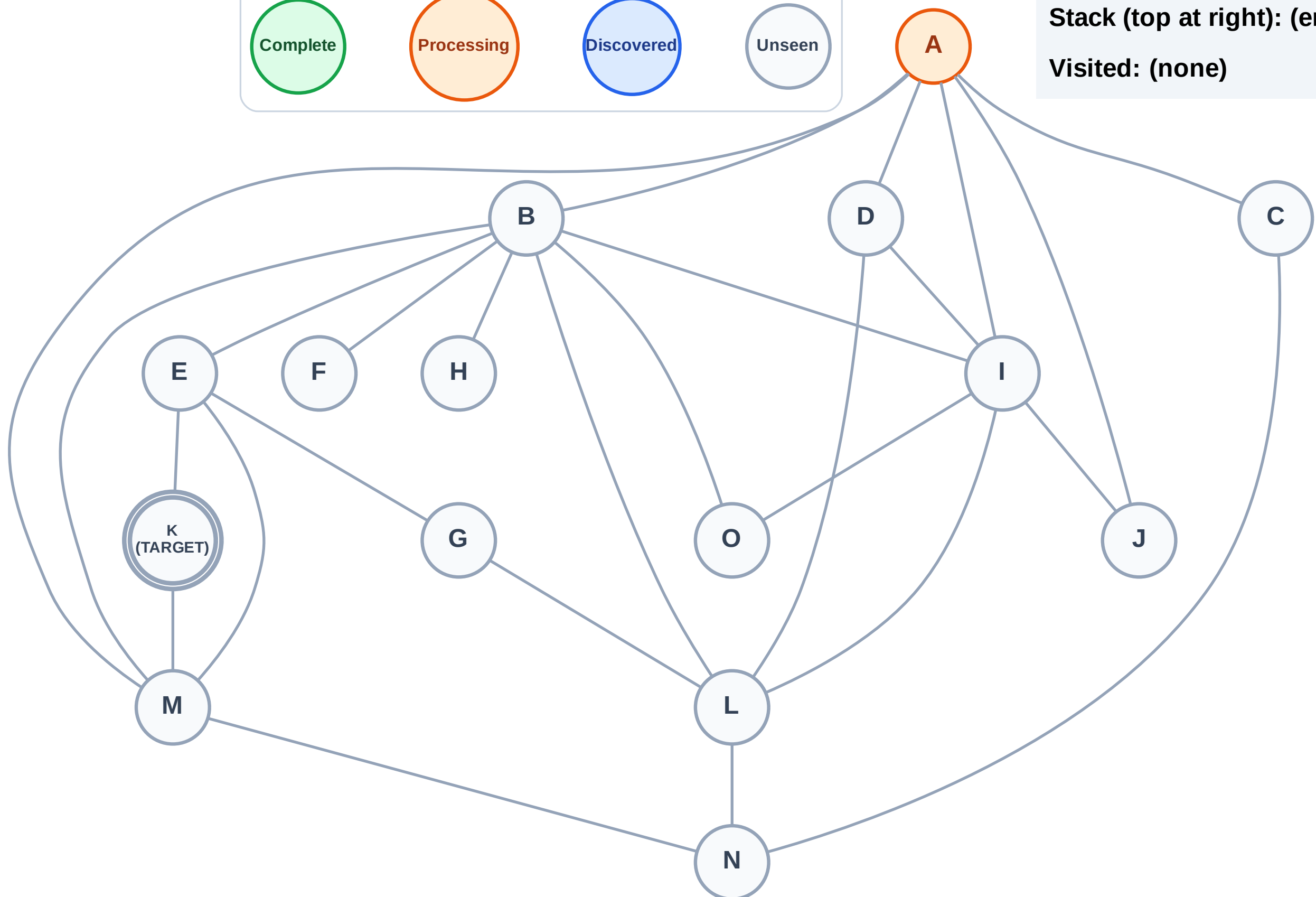
Discovered



Unseen

Stack (top at right): (empty)

Visited: (none)



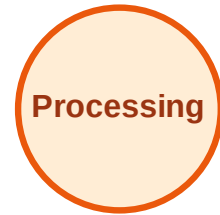
DFS Traversal

Step 3: Discover M, J, I, D, C, B from A

Legend



Complete



Processing



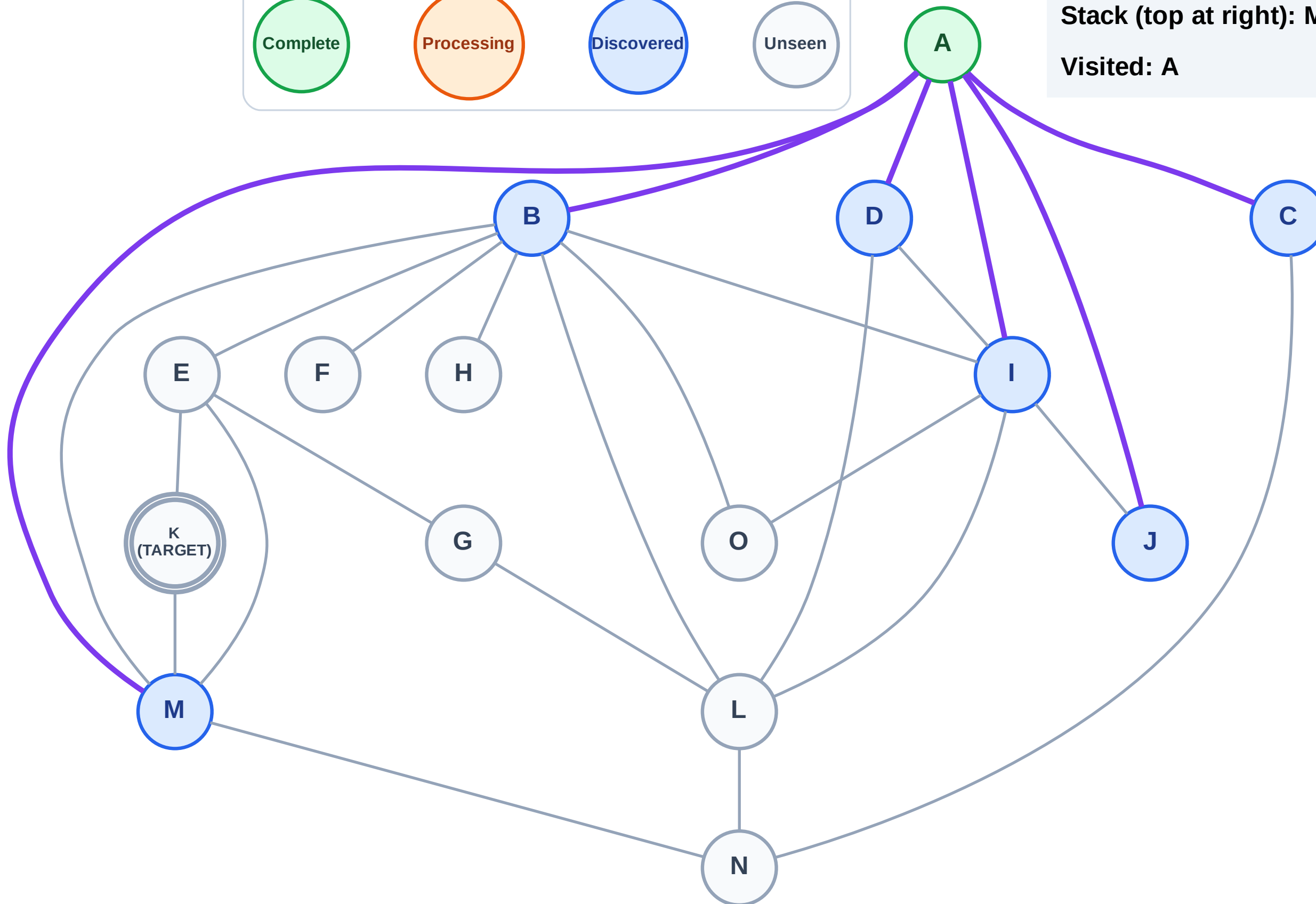
Discovered



Unseen

Stack (top at right): M | J | I | D | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

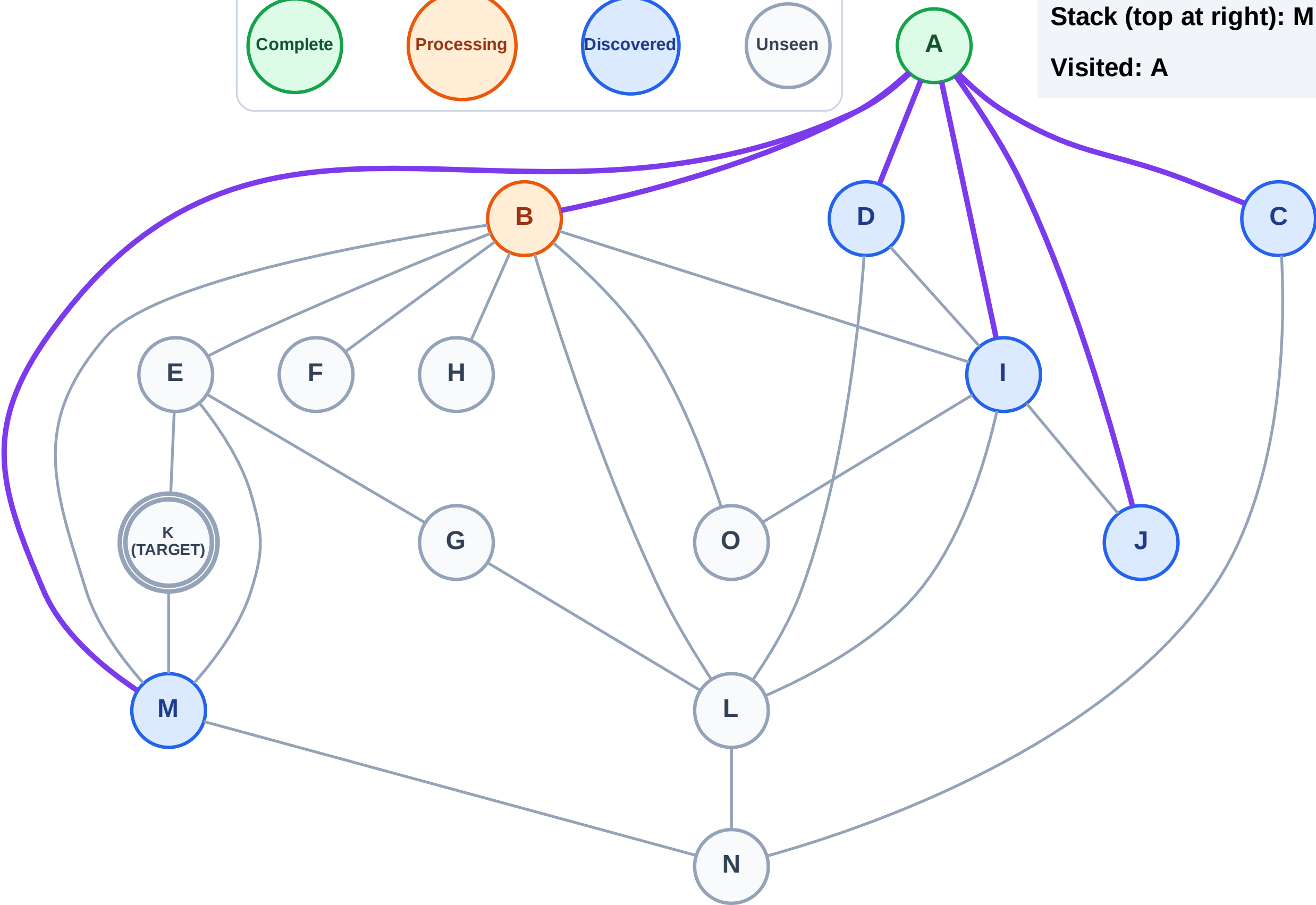
Processing

Discovered

Unseen

Stack (top at right): M | J | I | D | C

Visited: A



DFS Traversal

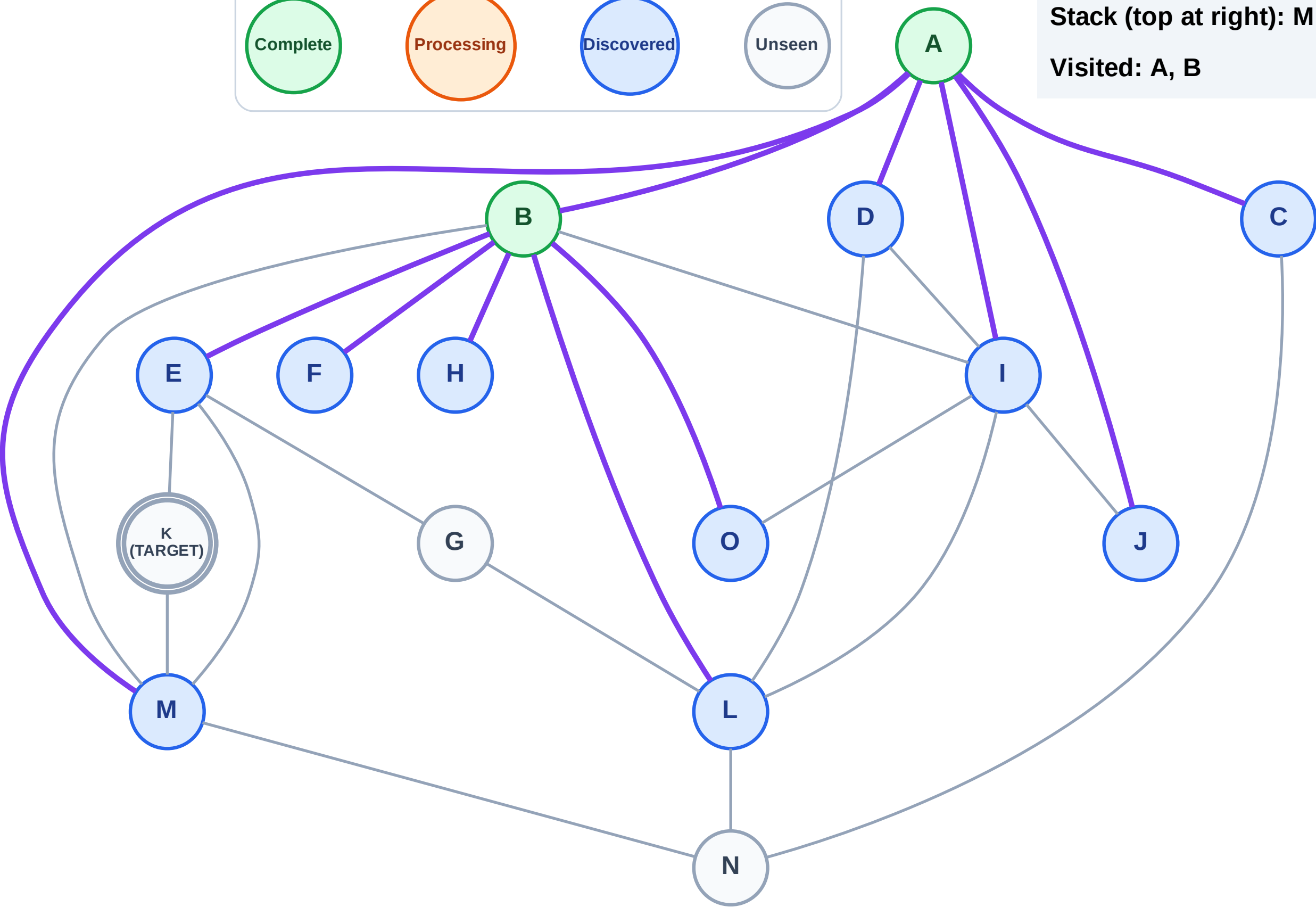
Step 5: Discover O, L, H, F, E from B

Legend



Stack (top at right): M | J | I | D | C | O | L | H | F | E

Visited: A, B



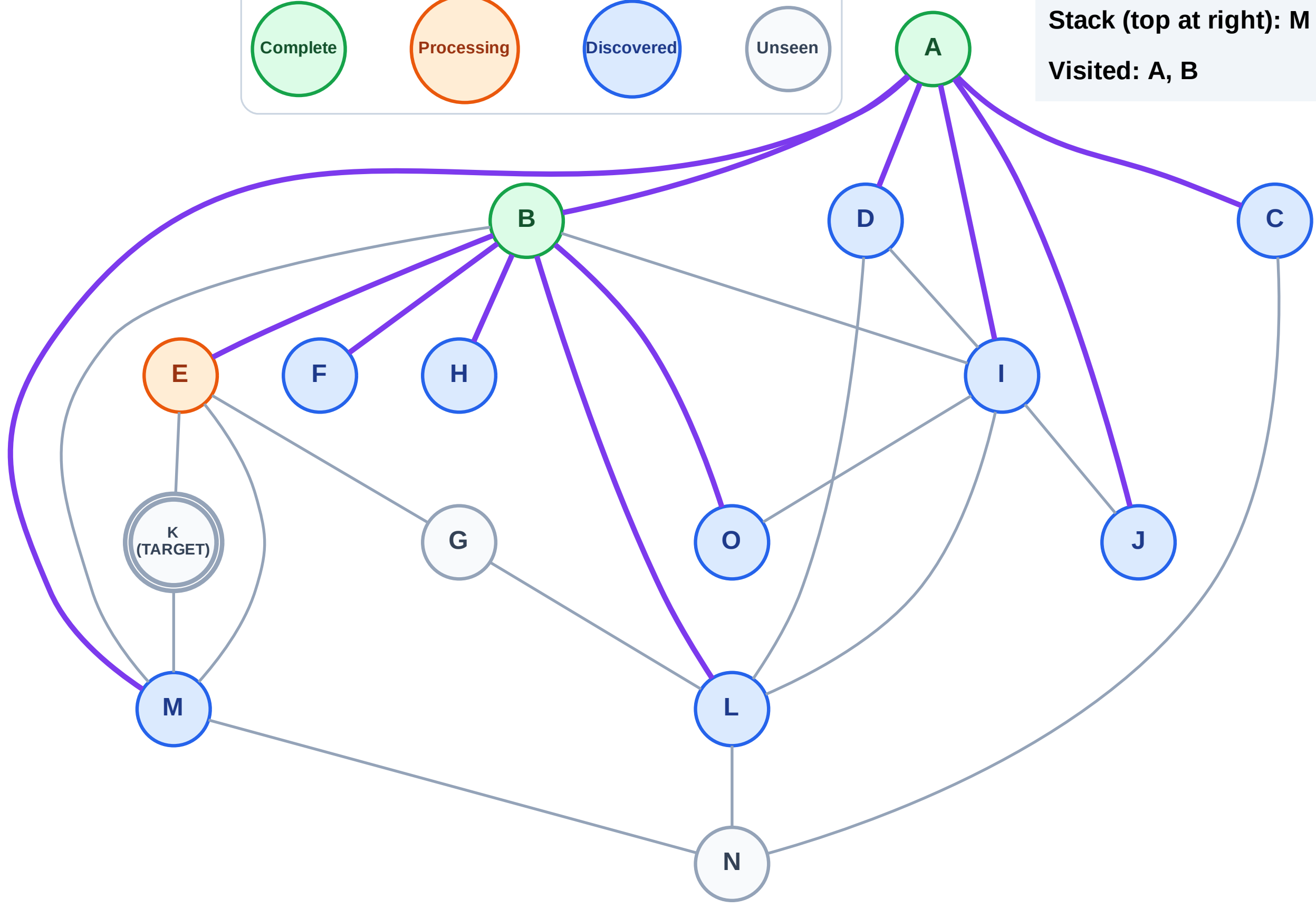
DFS Traversal

Step 6: Process node E

Legend



Stack (top at right): M | J | I | D | C | O | L | H | F
Visited: A, B



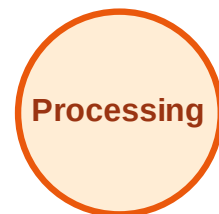
DFS Traversal

Step 7: Discover K, G from E

Legend



Complete



Processing



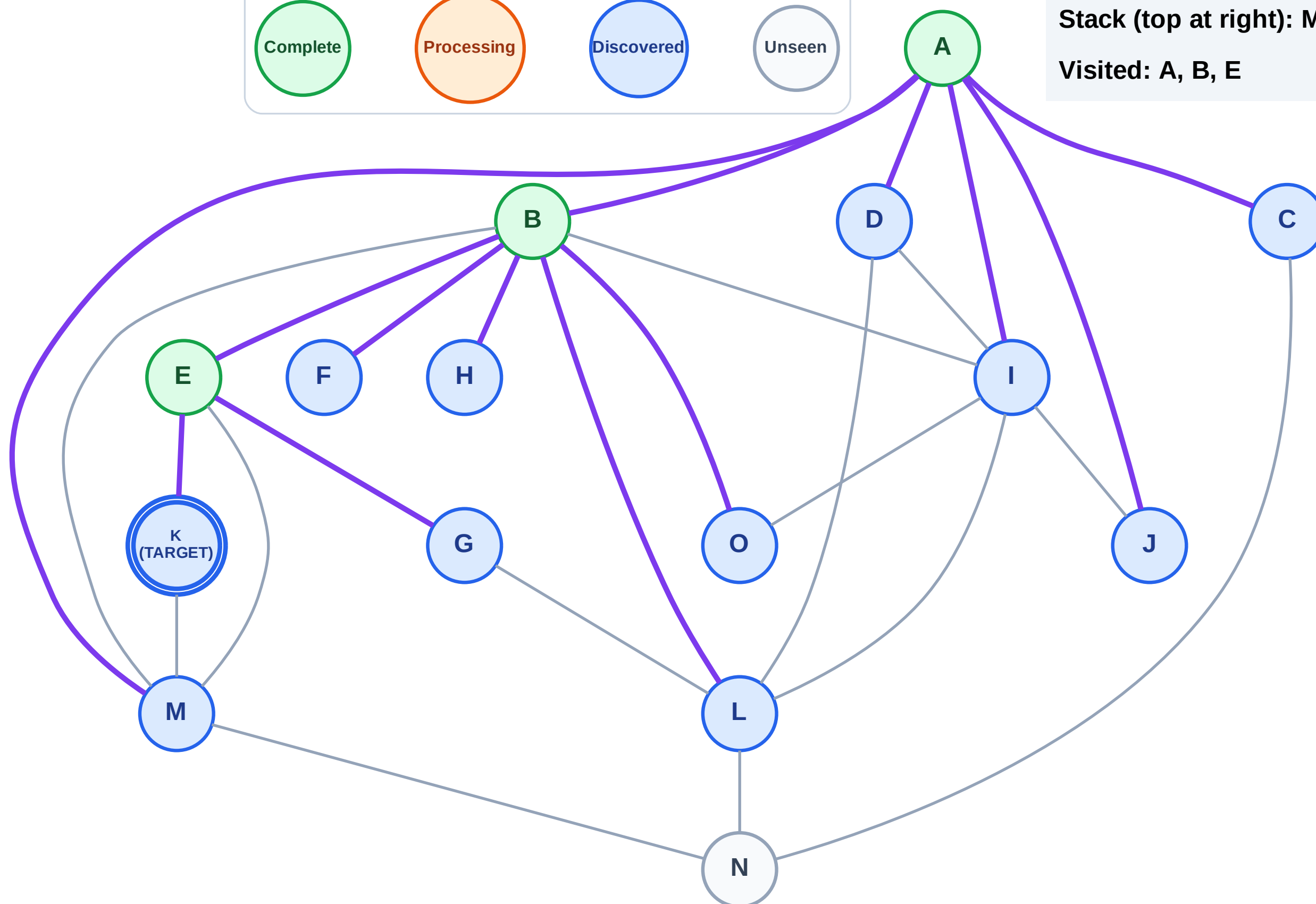
Discovered



Unseen

Stack (top at right): M | J | I | D | C | O | L | H | F | K | G

Visited: A, B, E



DFS Traversal

Step 8: Process node G

Legend

Complete

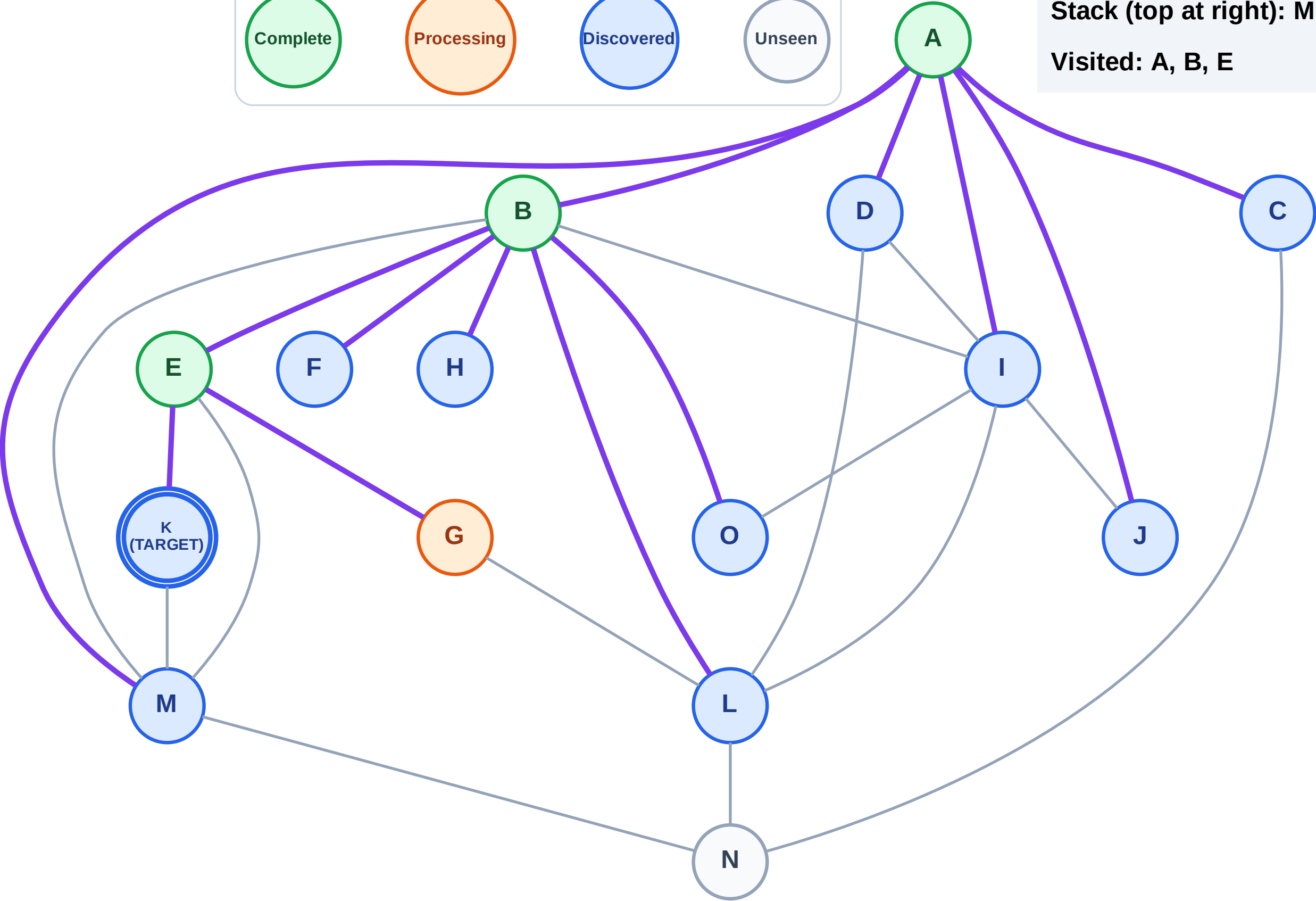
Processing

Discovered

Unseen

Stack (top at right): M | J | I | D | C | O | L | H | F | K

Visited: A, B, E



DFS Traversal

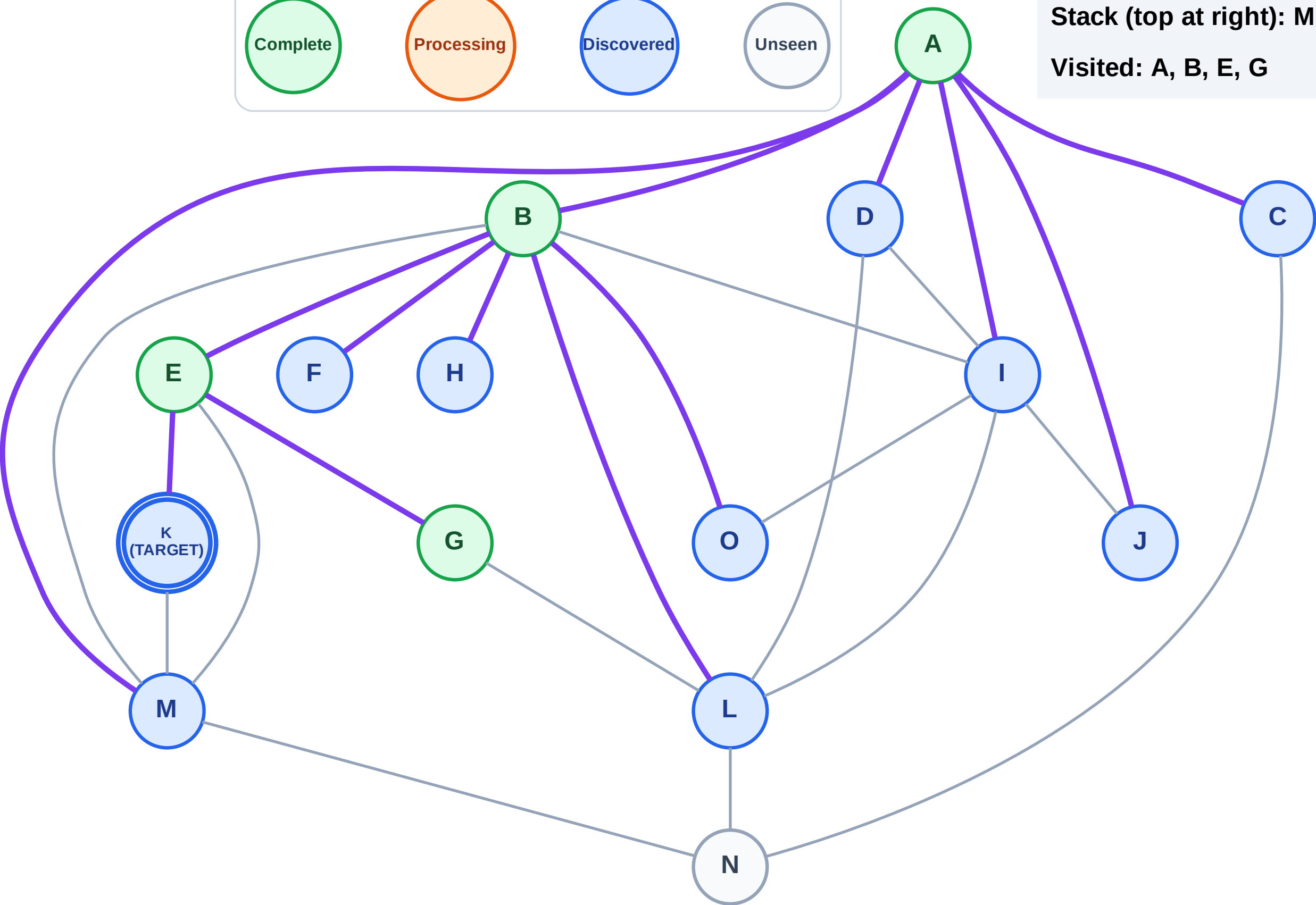
Step 9: No new neighbors from G

Legend



Stack (top at right): M | J | I | D | C | O | L | H | F | K

Visited: A, B, E, G



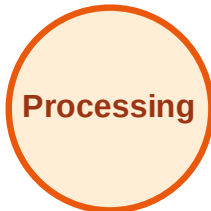
DFS Traversal

Step 10: Process node K

Legend



Complete



Processing



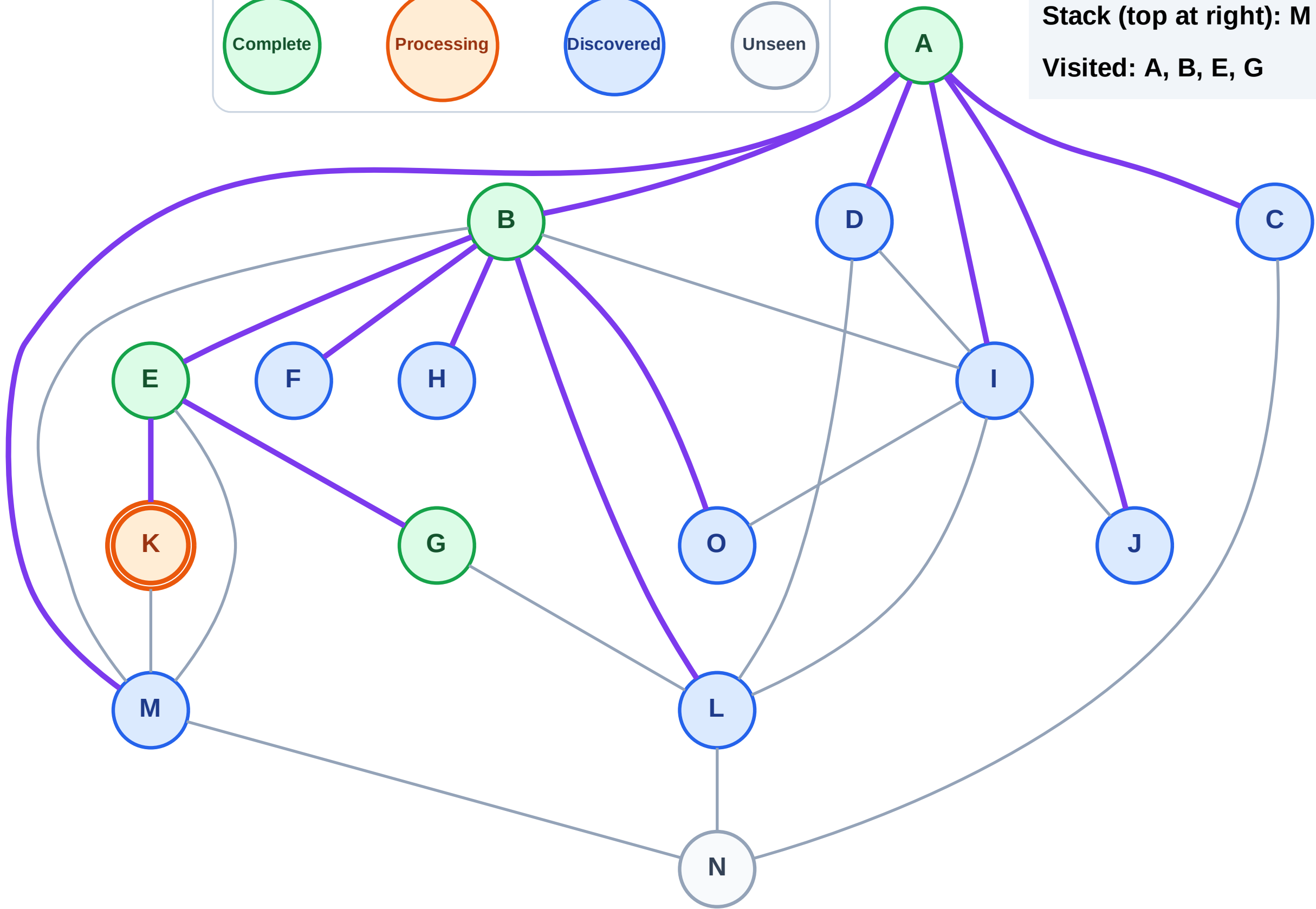
Discovered



Unseen

Stack (top at right): M | J | I | D | C | O | L | H | F

Visited: A, B, E, G



DFS Traversal

Step 11: TARGET NODE K REACHED! Path (3 hops): A → B → E → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): M | J | I | D | C | O | L | H | F

Visited: A, B, E, G, K

